

KROM and its Blind Spots

A unified analysis of kernelized reduced-order modeling and bifurcations:
which normal forms it handles, which defeat it, and the path through Moore–Greitzer

Research Note

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Abstract

We give a self-contained analysis of the KROM framework (Bacho, Lee, Owhadi, 2026) for kernelized reduced-order modeling of nonlinear PDEs and develop in detail its structural limitations. After laying out the optimal-recovery formulation, the empirical kernel built from snapshots, the sparse-Cholesky machinery, and the convergence theory, we identify the assumption that controls KROM’s behavior: a uniform-coercivity / analytic-parameter-dependence condition on the parametric solution map. We then build, step by step, a characterization of *which dynamical regimes KROM handles and which defeat it*, organized by the local normal form of the underlying dynamics. The characterization is then anchored by an extensive catalogue of PDE benchmarks — Moore–Greitzer with its five elementary local bifurcations (codim-1 stall Hopf, codim-2 mixed surge–stall double-Hopf, fold of axisymmetric equilibria, pitchfork on the rotation-symmetry-fixed slice, Bogdanov–Takens point in the throttle–plenum plane), Bratu–Gelfand, Swift–Hohenberg, the diffusive Brusselator, the complex Ginzburg–Landau equation, Kuramoto–Sivashinsky, Rayleigh–Bénard convection, and the nonlinear Schrödinger equation in a double-well potential. For each benchmark we identify the bifurcating object, the spectral / eigenvalue condition that locates it, and the explicit Kolmogorov-barrier obstruction that defeats KROM. From this catalogue we extract three universal failure modes: (F1) branch separation, (F2) continuous-symmetry curvature, and (F3) topological / unbounded-dimension growth. We then describe the kernel-based normal-form learning pipeline developed in parallel for the Moore–Greitzer compressor model as the principled response to (F1) at codim-1 Hopf, sketch how the same architecture extends to other codim-1 normal forms, and outline a unified branch-aware framework that combines KROM in coercive regions with normal-form patches across bifurcations. The report is intended to be self-contained: all relevant linear-stability calculations, normal-form data, and Kolmogorov N -width inequalities are stated and motivated, with intuition presented alongside the technical arguments.

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1 Introduction and roadmap

1.1 Background

Reduced-order modeling (ROM) of nonlinear partial differential equations is dominated by two families of techniques: linear projection methods (proper orthogonal decomposition, reduced-basis methods, dynamic mode decomposition) and nonlinear surrogate methods (autoencoders, neural operators, physics-informed neural networks). A more recent thread, originating in Owhadi’s program on kernel methods for PDEs and uncertainty quantification, casts PDE solution as a Gaussian-process optimal-recovery problem in a reproducing kernel Hilbert space (RKHS). The KROM method of Bacho, Lee, and Owhadi pushes this thread further by combining two ingredients:

- (i) an *empirical kernel* built from a snapshot library of PDE solutions, which biases the recovery toward the solution manifold;
- (ii) a *sparse Cholesky factorization* of the inverse kernel matrix, which exploits the screening effect from spatial statistics to reduce online complexity to near-linear.

The combination yields a non-intrusive, kernelized ROM that empirically outperforms generic Matérn-based GP solvers on a range of nonlinear PDEs (semilinear elliptic problems, discontinuous-coefficient Darcy flow, viscous Burgers, Allen–Cahn, 2D Navier–Stokes).

1.2 The question this report answers

A natural question, raised in conversations within this project, is: *can we say that KROM works for some systems and fails for others, in a way that is characterized by the local normal form of the dynamics?* The answer is yes, and the characterization can be made remarkably precise. This report builds it carefully.

The structure is as follows. Sections 2–3 review the KROM framework and isolate the structural assumption on which its convergence guarantees rest. Section 4 identifies what breaks at a bifurcation. Section 5 is the central new content: it characterizes the bifurcations and normal forms KROM can handle and provides a taxonomy for those it cannot, organized by codimension, branch multiplicity, and symmetry. Sections 6 and 7 instantiate the taxonomy on a comprehensive catalogue of PDE benchmarks — the Moore–Greitzer compressor model with its five elementary bifurcations, plus seven additional canonical PDEs spanning the bifurcation landscape. Section 8 extracts three universal failure modes from the catalogue. Section 9 compares POD and KROM at bifurcations in detail, showing that the two methods share the same approximation-theoretic blind spot but diverge mechanically. Section 10 describes the right replacement object, a learned nonlinear conjugacy onto a normal form. Section 11 shows how this is realized concretely for Moore–Greitzer. Section 12 sketches a unified branch-aware framework. Appendix A collects reference data for each benchmark.

1.3 What is new in this version

A previous version of this report focused on the KROM analysis and the Moore–Greitzer extension. This version adds four substantial components: (i) a detailed normal-form-based characterization of when KROM works and when it fails (Section 5), (ii) a catalogue of seven additional PDE benchmarks beyond Moore–Greitzer that exemplify each branch of the characterization (Section 7), (iii) explicit quantitative lower bounds on the snapshot-bias term that show the obstructions are geometric rather than convergence-rate (Sections 7–8, with three propositions giving N -independent lower bounds), and (iv) a detailed comparison of POD and KROM at bifurcations, showing that

both methods share the same approximation-theoretic blind spot but with mechanically different failure modes (Section 9). The intent is to provide a single self-contained reference that both an applied analyst and a numerical analyst can use to decide where KROM applies, where it does not, and what the right substitute is in each regime.

2 The KROM framework: optimal recovery on a snapshot RKHS

We summarize the KROM construction; the exposition follows the original paper, with notation slightly streamlined.

2.1 The PDE problem

Let $\Omega \subset \mathbb{R}^n$ be a Lipschitz domain, $T > 0$, and consider the initial–boundary value problem

$$\begin{cases} \mathcal{P}(t, x, u) = f, & x \in \Omega, \ t \in (0, T), \\ \mathcal{B}(t, x, u) = g, & x \in \partial\Omega, \ t \in (0, T), \\ \mathcal{I}(x, u) = h, & x \in \Omega, \end{cases} \quad (1)$$

where $\mathcal{P}$ is a (possibly nonlinear) differential operator, $\mathcal{B}$ a boundary operator, and $\mathcal{I}$ an initial-condition operator. Write $\tilde{\Omega} = [0, T] \times \Omega$ and $z = (t, x) \in \tilde{\Omega}$.

2.2 Optimal recovery in an RKHS

KROM picks a positive semidefinite kernel K on $\tilde{\Omega} \times \tilde{\Omega}$ with associated RKHS $\mathcal{H}_K$ and seeks the minimum-norm element of $\mathcal{H}_K$ that satisfies the PDE constraints at a finite collocation set:

$$\min_{v \in \mathcal{H}_K} \|v\|_K^2 \quad \text{s.t.} \quad \begin{cases} \mathcal{P}(z_i, v) = f(z_i), & i = 1, \dots, M_\Omega, \\ \mathcal{B}(\tilde{z}_j, v) = g(\tilde{z}_j), & j = 1, \dots, M_{\partial\Omega}, \\ \mathcal{I}(x_k, v) = h(x_k), & k = 1, \dots, L. \end{cases} \quad (2)$$

By the representer theorem, the minimizer admits the finite-dimensional form

$$u(z) = \sum_{m=1}^{N_{\text{data}}} c_m K(z, z_m), \quad N_{\text{data}} = DM_\Omega + M_{\partial\Omega} + L, \quad (3)$$

with coefficients chosen so that the constraints in (2) hold. For nonlinear $\mathcal{P}$, the coefficient system is solved by Gauss–Newton iteration; for linear operators it reduces to a single linear system.

2.3 Why the kernel choice matters

The kernel K fixes the RKHS $\mathcal{H}_K$ and therefore the notion of “simplicity” promoted by the minimum-norm principle. Generic stationary kernels (Matérn, Gaussian) encode global isotropic regularity, which is poorly suited to PDEs whose solutions exhibit transport-aligned features, localized layers, sharp transitions, or coherent structures with few effective degrees of freedom. KROM’s key move is to replace the generic kernel by a problem-adapted one.

2.4 The empirical kernel

Suppose we have a snapshot library $\{u_i\}_{i=1}^N$, obtained by varying forcing terms, boundary conditions, initial conditions, or parameters while keeping the operators and the domain fixed. The *empirical kernel* is

$$K_N(z, z') = \frac{1}{N} \sum_{i=1}^N u_i(z) u_i(z'), \quad z, z' \in \tilde{\Omega}. \quad (4)$$

This kernel has several equivalent interpretations:

- *Feature-map view.* Set $\Phi(z) = N^{-1/2}(u_1(z), \dots, u_N(z))$. Then $K_N(z, z') = \langle \Phi(z), \Phi(z') \rangle_{\mathbb{R}^N}$, so K_N is a linear kernel in a data-driven feature space whose coordinates are physically meaningful snapshot values.
- *Empirical-covariance view.* If snapshots are interpreted as samples from the distribution of solutions relevant to the task, then K_N is the empirical covariance of the random field $u(z)$, and the minimum-norm interpolant becomes the noiseless GP posterior mean under the corresponding prior.
- *ROM view.* The induced RKHS is exactly the snapshot span,

$$\mathcal{H}_{K_N} = \text{span}\{u_1, \dots, u_N\},$$

with a data-dependent inner product. The recovered solution is therefore a linear combination of snapshots, and KROM is a kernelized packaging of POD/reduced-basis methods.

The third interpretation is the crucial one for what follows: *KROM with the empirical kernel approximates the new solution by a linear combination of training snapshots.* This linearity is the source of every blind spot we will identify.

2.5 Homogeneous linear constraints satisfied by construction

If $\mathcal{B}$ is linear and all snapshots satisfy a homogeneous boundary condition $\mathcal{B}u_i = 0$, then any linear combination $\sum_i \beta_i u_i$ satisfies the same condition, so this constraint need not be enforced explicitly in the collocation system. Inhomogeneous constraints can be reduced to the homogeneous case by lifting.

2.6 The sparse Cholesky factorization

The cost bottleneck of (2) is the dense kernel matrix $K(\phi, \phi) \in \mathbb{R}^{N_{\text{data}} \times N_{\text{data}}}$. KROM uses the sparse Cholesky algorithm of Schäfer–Katzfuss–Owhadi to approximate the Cholesky factor of $K(\phi, \phi)^{-1}$ to within Frobenius error ε in

$$\mathcal{O}\left(N_{\text{data}} \log^d(N_{\text{data}}/\varepsilon)\right) \text{ space}, \quad \mathcal{O}\left(N_{\text{data}} \log^{2d}(N_{\text{data}}/\varepsilon)\right) \text{ time}.$$

The algorithm rests on a *maximin reordering* of the collocation points,

$$i_{q+1} = \arg \max_{i \in \mathcal{I} \setminus \{i_1, \dots, i_q\}} \text{dist}(x_i, \{x_{i_1}, \dots, x_{i_q}\}), \quad (5)$$

which yields a geometric covering of Ω with cells of diameter $\sim cj^{-1/d}$ after j steps. Under this reordering, the off-diagonal entries of the Cholesky factor of K^{-1} exhibit exponential decay because of the screening effect: conditioning on values at a sufficiently dense intermediate set makes points

in distant regions approximately conditionally independent. The matrix sparsity is controlled by a single parameter $\rho > 0$, with the nonzero pattern

$$S_{P,\ell,\rho} = \{(i, j) : i \leq j, \|x_{P(i)} - x_{P(j)}\| \leq \rho \ell_j\},$$

where ℓ_j is a length scale tied to the maximin distance at step j . Importantly, the screening effect is a statement about *spatial* conditional covariance, governed by the Euclidean metric on Ω , which has no a priori connection to the dynamical structure of the PDE.

2.7 Putting it together

Online, KROM solves (2) with the empirical kernel (4), accelerated by the sparse Cholesky factorization. Offline, one constructs the snapshot library and the kernel matrix once. Empirically, the method matches or outperforms Matérn-2.5 baselines across a broad range of nonlinear PDEs. Crucially for our analysis below, the entire architecture treats the solution as living in a *linear* subspace $\mathcal{H}_{K_N} = \text{span}\{u_1, \dots, u_N\}$ of the ambient function space, with ρ controlling spatial-statistical screening, not dynamical structure.

3 The KROM error analysis: where the assumptions live

The convergence theory of KROM separates the total error into three contributions. Understanding where each one lives is essential for diagnosing the method's blind spots.

3.1 The two-term error estimate

Let u^* be the true solution of (1), and $u_{N,h}^\dagger \in \mathcal{H}_{K_N}$ the KROM minimizer with collocation set of fill distance $h_\Omega = \sup_{x_0 \in \Omega} \inf_{x \in X_\Omega} \|x_0 - x\|$. The main result of the paper is the two-term bound

$$\left\| u_{N,h}^\dagger - u^* \right\|_{H^\ell(\Omega)} \leq Ch_\Omega^\gamma \|u_N^*\|_{\mathcal{H}_{K_N}} + \inf_{v \in \mathcal{H}_{K_N}} \|v - u^*\|_{H^\ell(\Omega)}, \quad (6)$$

where u_N^* is any element of $\mathcal{H}_{K_N}$ satisfying the continuous PDE constraints and $\gamma = s - \ell > 0$ is determined by RKHS regularity. The decomposition is illuminating:

- The first term, $Ch_\Omega^\gamma \|u_N^*\|_{\mathcal{H}_{K_N}}$, is a *discretization / fill-distance error* coming from finite collocation. It vanishes as $h_\Omega \rightarrow 0$ at rate γ , provided $\|u_N^*\|_{\mathcal{H}_{K_N}}$ stays bounded.
- The second term, $\inf_{v \in \mathcal{H}_{K_N}} \|v - u^*\|_{H^\ell(\Omega)}$, is the *snapshot-space bias*: the best possible H^ℓ -approximation of the truth by an element of the snapshot span.

The sparse-Cholesky approximation contributes an additional error of order ε that can be made arbitrarily small by tuning ρ , and which the paper conjectures decays exponentially in ρ for the empirical kernel (proven for Matérn). Of these three error sources, only the snapshot-space bias depends on the geometry of the solution set; the others are purely numerical.

3.2 The Kolmogorov N -width

The snapshot-space bias has a clean interpretation. If

$$\mathcal{M} = \{u(\mu) : \mu \in \mathcal{D}\} \subset V$$

is the solution manifold over the parameter set $\mathcal{D}$, with V a Hilbert space, then the best worst-case approximation by any N -dimensional subspace is the Kolmogorov N -width

$$d_N(\mathcal{M}; V) := \inf_{\substack{W \subset V \\ \dim W = N}} \sup_{u \in \mathcal{M}} \inf_{v \in W} \|u - v\|_V. \quad (7)$$

For a well-chosen snapshot space $\mathcal{H}_{K_N}$, the bias $\inf_{v \in \mathcal{H}_{K_N}} \|v - u^*\|_V$ is an instance of $d_N(\mathcal{M}; V)$.

3.3 The structural assumption that drives sub-exponential decay

The favorable rate of the snapshot-bias term rests on a specific assumption about the solution manifold. Following the paper, suppose the variational form is

$$a(u(\mu), v; \mu) = f(v; \mu), \quad \forall v \in V, \mu \in \mathcal{D},$$

and assume:

- (1) *Uniform coercivity and boundedness.* There exist $\alpha, \gamma > 0$ such that for all $\mu \in \mathcal{D}$,

$$\alpha \|v\|_V^2 \leq a(v, v; \mu), \quad |a(w, v; \mu)| \leq \gamma \|w\|_V \|v\|_V, \quad \forall v, w \in V.$$

- (2) *Affine and analytic parameter dependence.* The bilinear form and the data admit affine expansions whose coefficients extend analytically to a complex neighborhood of $\mathcal{D}$.

Under these hypotheses, the solution manifold $\mathcal{M}$ is compact and analytically parametrized in V , and the Kolmogorov N -width decays sub-exponentially:

$$d_N(\mathcal{M}; V) \leq C \exp(-cN^\alpha), \quad N \geq 1, \quad (8)$$

for some $c > 0$ and $\alpha \in (0, 1]$ depending on the parameter dimension. For single-parameter problems one typically has $\alpha = 1$ and pure exponential decay; for P -parameter problems one obtains $\alpha = 1/P$ up to logarithmic factors. *This is the structural reason why KROM's snapshot-bias term decays well in practice on smooth-parameter problems, and the structural reason why it fails when those hypotheses break.*

4 The bifurcation obstruction: what actually breaks

We now turn to the central question: does KROM still work when the PDE undergoes a bifurcation? We argue carefully that it does not. The obstruction lives in the snapshot-bias term of the error estimate (6) and is rooted in the structural assumptions that drive the favorable Kolmogorov N -width decay of Section 3.3.

4.1 Coercivity and analyticity break at a bifurcation point

The favorable Kolmogorov N -width decay (8) relies on Assumption 3.3. Both of its components fail at a bifurcation point, as a matter of definition.

Failure of uniform coercivity. For a parametrized linear elliptic problem $L(\mu)u = f(\mu)$, the variational form $a(u, v; \mu) = (L(\mu)u, v)$ is uniformly coercive only if $L(\mu)$ is uniformly invertible across $\mathcal{D}$. At a bifurcation parameter μ_b , by definition, the linearization $DG(u^*(\mu_b); \mu_b)$ has an eigenvalue on the imaginary axis (Hopf), or zero (saddle-node, pitchfork, transcritical). The bilinear form associated with the linearization develops a nontrivial kernel, and uniform coercivity in μ across any neighborhood of μ_b fails. For nonlinear problems the same failure manifests at the level of the linearized operator that governs continuation along the branch.

Failure of analytic parameter dependence. Across a bifurcation, the solution manifold $\mathcal{M} = \{u(\mu) : \mu \in \mathcal{D}\}$ is no longer a single analytic graph over $\mathcal{D}$. At a pitchfork, the manifold acquires a new branch; at a saddle-node, the parametrization fails (vertical tangent in μ); at a Hopf bifurcation, a new family of periodic orbits with amplitude $\sim |\mu - \mu_b|^{1/2}$ emerges and the solution map $\mu \mapsto u(\mu)$ is no longer analytic at μ_b if one regards limit cycles as part of the set of long-time behaviors. The affine analytic parameter dependence required by Assumption 3.3 is incompatible with this picture.

4.2 The Kolmogorov N -width breakdown

Once the structural assumptions break, the conclusion (8) can no longer be invoked. More importantly, the geometric reason *why* the conclusion fails is exactly the reason KROM is a poor choice in this regime.

Across a bifurcation, the solution manifold $\mathcal{M}$ may be:

- *Multi-sheeted.* A pitchfork at μ_b produces, for $\mu > \mu_b$, two distinct stable equilibria $u_+(\mu)$ and $u_-(\mu)$. A Hopf at μ_b replaces a stable equilibrium by an unstable equilibrium plus a limit cycle, so the natural “solutions of physical interest” include both an equilibrium and a 1-parameter family of points on the cycle.
- *Non-smooth in μ .* The amplitude of the bifurcating limit cycle in a Hopf bifurcation grows as $\sqrt{\mu - \mu_b}$, which is Hölder but not Lipschitz. The map $\mu \mapsto u(\mu)$ has no analytic extension across μ_b .

Linear approximation by an N -dimensional subspace works well precisely for manifolds that are smooth, single-sheeted, and uniformly close to flat. Multi-sheeted, non-smooth manifolds defeat linear N -widths: the lower bound

$$d_N(\mathcal{M}; V) \geq \frac{1}{2} \text{dist}_V(\text{branch A}, \text{branch B})$$

holds eventually for two well-separated branches, regardless of N . Concretely, snapshots collected from one branch cannot, by linear combination, reach a solution on the other branch.

4.3 The empirical kernel inherits the obstruction

KROM’s empirical-kernel RKHS is exactly the linear span of snapshots,

$$\mathcal{H}_{K_N} = \text{span}\{u_1, \dots, u_N\}.$$

If the snapshots are drawn from a single branch of $\mathcal{M}$, then $\mathcal{H}_{K_N}$ is contained in a neighborhood of that branch, and

$$\inf_{v \in \mathcal{H}_{K_N}} \|v - u^*\|_{H^\ell(\Omega)} \geq \text{dist}_{H^\ell}(u^*, \text{branch A})$$

whenever u^* lives on a different branch. The snapshot-bias term in (6) is then bounded below by a quantity that does not decay in N .

If snapshots are drawn from both branches, the snapshot span overshoots: linear combinations $\beta_+ u_+ + \beta_- u_-$ produce intermediate states that lie on neither branch and that do not in general satisfy the PDE. The minimum-norm principle tries to interpolate between branches with no guarantee that the intermediate function is dynamically meaningful, and Gauss–Newton iterations at a snapshot-blended initial guess may either stall or be attracted to a spurious solution.

4.4 The Gauss–Newton singularity at the bifurcation

There is a second, separate obstruction that is operational rather than approximation-theoretic. KROM solves the nonlinear system (2) by Gauss–Newton iteration. The Newton system is

$$\mathcal{L}(\mu)\delta u = -[\mathcal{P}(u^{(k)}) - f],$$

and at a bifurcation parameter $\mathcal{L}(\mu_b)$ has the critical eigenmode ϕ^c in its kernel. This is a generic issue at any fold or pitchfork; it is well-known that one needs pseudo-arclength continuation or tangent-following to circumvent it. KROM as currently formulated does not implement such continuation, so it cannot pass through a fold even when the snapshot library is rich enough to span both branches separately.

4.5 Sparse Cholesky does not help

The sparse-Cholesky / maximin ingredient of KROM is orthogonal to all of the obstructions above. The screening-effect decay of conditional covariance is governed by the spatial geometry of the collocation points and by smoothness of the kernel along the spatial domain Ω . It tells us nothing about whether the kernel K_N is well-suited to represent the new solution. If the snapshot library spans the wrong portion of solution space, no amount of sparsification can fix the snapshot-bias term in (6).

4.6 Summary: KROM’s blind spot, precisely stated

Claim. KROM with an empirical kernel built from a snapshot library on a single branch of the solution manifold fails as the parameter crosses a bifurcation point, because (i) the uniform-coercivity / analytic-parameter-dependence assumption that drives sub-exponential Kolmogorov N -width decay breaks at the bifurcation, (ii) the solution manifold becomes multi-sheeted or non-smooth in the parameter, (iii) the snapshot-span RKHS cannot, by linear combination, reach solutions on another branch, and (iv) the Gauss–Newton iteration is singular at the bifurcation point. The failure is a manifold-approximation / Kolmogorov-barrier issue. The sparse-Cholesky machinery is unaffected by this obstruction but also cannot fix it.

5 Characterization: which normal forms KROM handles, and which defeat it

We now turn to the central new content of this report: a precise, normal-form-based characterization of when KROM works and when it fails. We give first the cleanest version (as a single statement), then break it down by codimension and unfolding type, with intuition and quantitative consequences for each case.

5.1 The clean version

The characterization can be stated as a single dichotomy.

Characterization. KROM with an empirical kernel works exactly when the relevant portion of the solution manifold is the graph of a single analytic function $\mu \mapsto u(\mu)$ over the parameter set, with uniformly invertible linearization. Equivalently, in normal-form

language: *KROM works precisely when the local normal form of the parametrized dynamics has empty critical spectrum on the parameter region of interest.* Every eigenvalue of the linearization stays in the open left or right half-plane uniformly across $\mathcal{D}$. The moment any eigenvalue touches the imaginary axis on $\mathcal{D}$, KROM fails.

The reason is exactly the analysis of Section 4: empty critical spectrum is equivalent to uniform hyperbolicity along the equilibrium branch, which is equivalent to uniform coercivity of the linearized variational form, which is precisely the structural assumption that drives the sub-exponential Kolmogorov N -width decay. Conversely, any normal form with nonempty critical part introduces multi-sheetedness or non-smoothness into $\mathcal{M}$ that defeats linear approximation.

This dichotomy is uniform across normal-form types — it is not the case that some bifurcations are harmless and others fatal. Every codim- k normal form with $k \geq 1$ becomes a problem if the parameter set crosses the bifurcation locus.

5.2 A finer-grained taxonomy: severity of failure by normal form

Although every bifurcation crossing breaks KROM in principle, in practice the severity varies considerably and is genuinely characterized by the normal form. The relevant axis is the *branch multiplicity* that the bifurcation introduces into $\mathcal{M}$: how many disjoint sheets coexist on the relevant side of the bifurcation, what dynamical objects (equilibria, cycles, tori, homoclinics) coexist there, and how the topology of the long-time attractor changes. We collect the picture in Table 1.

Table 1: Severity of KROM failure ordered by normal-form type. The severity is measured by branch multiplicity, dynamical-object count, and how rapidly the required snapshot library size grows with the parameter region considered.

Normal form	What happens to $\mathcal{M}$	KROM failure mode
Saddle-node $\dot{x} = \mu - x^2$	Branch terminates at a fold; $\mu \mapsto u(\mu)$ has vertical tangent at μ_b	<i>Mild.</i> Local non-analyticity (square-root scaling) on one side; no new sheet on either side. Snapshots from one side cannot continue past the fold. Gauss–Newton singular at the fold itself. Recoverable by pseudo-arclength continuation of the snapshot library.
Transcritical $\dot{x} = \mu x - x^2$	Two branches cross transversally and exchange stability at μ_b	<i>Moderate.</i> Two sheets coexist on both sides. Linear blends are spurious near the crossing. Snapshots from both sides can be collected, but the kernel does not respect the discrete branch label, so KROM may resolve a spurious mid-branch state.
Pitchfork $\dot{x} = \mu x - x^3$	One branch becomes three (one unstable, two stable mirror images)	<i>Severe.</i> Multi-sheeted post-bifurcation. The post-pitchfork manifold is non-flat in any subspace containing both stable branches: $u_+ + u_- = 2u_b$ (the unstable branch), so the linear combination crosses through the unstable equilibrium. Snapshot library would need to span all three branches and would still produce convex combinations that lie off-manifold.

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Table 1 continued from previous page

Normal form	What happens to $\mathcal{M}$	KROM failure mode
Hopf $\dot{z} = (\rho + i\omega)z + \beta \ z\ ^2 z$	Equilibrium branch persists but a 1-parameter family of limit cycles is born	<i>Severe</i> . The limit cycle is bounded away in the natural Hilbert norm from the equilibrium, with separation $\sim \sqrt{ \mu - \mu_b }$. No linear combination of equilibrium snapshots reaches the cycle. If cycle snapshots are also added, the linear span produces inadmissible quasi-equilibrium intermediate states.
Bautin (degenerate Hopf)	Locus where first Lyapunov coefficient β_0 vanishes; super- and sub-critical Hopf branches meet	<i>Severe</i> (codim-2). Two cycle families collide; need quintic normal form. Critical Lyapunov coefficient is the relevant invariant.
Bogdanov–Takens	Equilibria, limit cycles, and homoclinic orbits all in a 2-parameter unfolding	<i>Catastrophic</i> (codim-2). Three distinct dynamical objects coexist, all needed for a complete reduced model; one of them (the homoclinic orbit) has no Taylor normal form at all.
Double-Hopf (1:1, 1:2, non-resonant)	4D center manifold with possibly invariant tori, mode interactions, secondary Hopf	<i>Catastrophic</i> (codim-2). Local center-manifold dimension is 4; $\mathcal{M}$ may include tori; snapshot library has wrong intrinsic dimension.
Steady-state O(2)-symmetric pitchfork	Pitchfork in a continuous-symmetry setting; bifurcating set is a circle of solutions	<i>Severe</i> but symmetric. The bifurcating manifold is a non-flat circle inside a 2D linear span; KROM hits failure mode (F2) of Section 8.
Period-doubling cascade	Sequence of cycles of doubling period	<i>Severe</i> (cascade). At each step, one branch becomes two with disjoint period structure. Required snapshot count grows exponentially.
Cascade to chaos	Sequence of bifurcations of unbounded number, terminating in chaotic attractor	<i>Catastrophic</i> . Snapshot library size grows at least linearly with parameter, defeating the entire point of reduction (failure mode (F3)).

5.3 Pre- vs. post-bifurcation regions: an asymmetry

There is a useful asymmetry that often gets glossed over. Take a supercritical Hopf at μ_H :

- For $\mu > \mu_H$ (pre-bifurcation, equilibrium stable): KROM works fine. The branch is uniformly hyperbolic on any closed interval bounded away from μ_H . Snapshots from this region span the equilibrium branch, $\mathcal{M}$ is a smooth graph, N -width decays exponentially.
- For $\mu < \mu_H$ (post-bifurcation, limit cycle exists): KROM still works *if* you only care about the unstable equilibrium and your snapshots track the equilibrium branch. But if you want to capture the limit cycle, you need it in your snapshot library, and now the snapshot manifold has higher local dimension (1 for equilibrium + 1 for the cycle phase). Worse, the cycle’s amplitude scales as $\sqrt{\mu_H - \mu}$, which is non-analytic at μ_H .

So even for the same bifurcation, KROM’s success depends on what the user wants to recover and how the parameter region is constrained. The clean characterization at the start of this section — “uniform hyperbolicity over $\mathcal{D}$ ” — is the requirement for a single reduced model that works everywhere on $\mathcal{D}$, which is the most common ROM use case.

5.4 Intuitive summary

Putting the technical and the intuitive together: *KROM is a smart linear interpolator in function space*. It works extremely well when the truth is a smooth function of parameters, even when that smooth function is highly oscillatory, has sharp gradients, lives on a discontinuous-coefficient elliptic problem, etc. What it cannot do is interpolate *across topological discontinuities* in the solution set. Bifurcations are exactly such discontinuities: the topology of the long-time behavior changes at μ_b , the solution set splits or merges, and no linear combination of pre-bifurcation snapshots can produce a post-bifurcation object that did not exist before.

The normal form is just the local mathematical name for what kind of topological discontinuity occurs and how violent it is. “Saddle-node” is the gentlest discontinuity (a branch ends); “Hopf” is moderately violent (a new periodic object is born from nothing); “double-Hopf” is more violent still (a four-dimensional invariant set is born); “cascade to chaos” is the most violent (an entire sequence of new dynamical structures is born one after another).

6 Catalogue, part I: the Moore–Greitzer compressor model

We turn now to making the characterization concrete on a series of PDE benchmarks. The Moore–Greitzer compressor model is the central example because, as a single PDE–ODE system, it exhibits all five elementary local bifurcations of finite-dimensional dynamical-systems theory. We describe the equations and then walk through each bifurcation, identifying the normal form, the bifurcation locus, and the manner in which KROM fails.

6.1 The model

The full viscous Moore–Greitzer model describes the onset of rotating stall and surge in axial-flow compressors. Its state is the triple

$$X(t) = (\Phi(t), \Psi(t), g(\cdot, t)) \in \mathbb{R} \times \mathbb{R} \times H_{\text{per}}^1(S^1) =: \mathcal{X},$$

where Φ is the (axisymmetric) mean flow coefficient, Ψ is the plenum pressure rise, and $g(\theta, t)$ is the circumferential disturbance about the mean. The dynamics couple a PDE for g on the circle to two ODEs for (Φ, Ψ) :

$$\begin{cases} \dot{\Phi} = F_{\Phi}(\Phi, \Psi, g; \gamma), \\ \dot{\Psi} = F_{\Psi}(\Phi, \Psi; \gamma), \\ \partial_t g = \mathcal{L}_{\gamma} g + \mathcal{N}_{\gamma}(\Phi, g), \end{cases} \quad (9)$$

where γ is the throttle parameter, $\mathcal{L}_{\gamma}$ is the linear convection–diffusion operator on the annulus, and $\mathcal{N}_{\gamma}$ encodes the compressor characteristic

$$\mathcal{N}_{\gamma}(\Phi, g) = \psi_c(\Phi + g) - \overline{\psi_c(\Phi + g)},$$

with ψ_c a typically cubic compressor characteristic and $\overline{\cdot}$ the angular average over $\theta \in [0, 2\pi)$.

There is a smooth axisymmetric equilibrium branch $X^*(\gamma) = (\Phi^*(\gamma), \Psi^*(\gamma), 0)$ parametrized by the throttle. As γ varies, this equilibrium passes through five distinct local bifurcations, each of which can be the focus of a different reduced-order modeling exercise.

6.2 Codim-1: stall Hopf

As γ decreases past a critical value γ_H , the first non-axisymmetric Fourier mode of g destabilizes, with a complex-conjugate eigenvalue pair $\lambda_{\pm} = \pm i\omega_H$ crossing the imaginary axis transversally. This is a generic codim-1 Hopf bifurcation; the local normal form on the 2D center manifold is

$$\dot{z} = (\rho(\gamma) + i\omega(\gamma))z + \beta(\gamma) \|z\|^2 z + \mathcal{O}(\|z\|^5),$$

with $\rho(\gamma_H) = 0$, $\omega(\gamma_H) = \omega_H$, and $\beta(\gamma_H)$ the first Lyapunov coefficient (positive in the supercritical case relevant to MG stall).

Why KROM fails. For $\gamma < \gamma_H$ the long-time attractor is no longer the axisymmetric equilibrium but a limit cycle in $\mathcal{X}$ corresponding to a rotating stall pattern. The cycle is bounded away in H^1 -norm from the equilibrium for any γ at which both exist. A snapshot library of equilibrium states from $\gamma > \gamma_H$ cannot, by linear combination, produce the rotating-stall limit cycle. Conversely, snapshots from $\gamma < \gamma_H$ lie near the cycle and cannot reach the equilibrium for $\gamma > \gamma_H$. There is no choice of snapshot library that escapes the dichotomy.

6.3 Codim-2: mixed surge–stall double-Hopf

When γ and the Greitzer B parameter are tuned simultaneously to a codim-2 locus, both the stall mode and the surge (axisymmetric oscillation) mode become critical at once. The local center manifold is 4D, and the normal form is a coupled-amplitude double-Hopf system

$$\dot{z}_s = (\rho_s + i\omega_s)z_s + a_{ss}z_s \|z_s\|^2 + a_{sg}z_s \|z_g\|^2 + \dots, \quad \dot{z}_g = (\rho_g + i\omega_g)z_g + a_{gs}z_g \|z_s\|^2 + a_{gg}z_g \|z_g\|^2 + \dots.$$

Why KROM fails. The local invariant set near the double-Hopf includes (depending on the cross-coupling coefficients a_{sg}, a_{gs}) a stable equilibrium, a stall-only cycle, a surge-only cycle, and a 2-torus combining both. A snapshot library that does not span all four objects cannot resolve the local dynamics; even if it does, the union is not a 4D manifold but a stratified set in $\mathcal{X}$, which $\mathcal{H}_{K_N}$ cannot represent.

6.4 Codim-1: fold of axisymmetric equilibria

The axisymmetric equilibrium condition $\psi_c(\Phi^*) = (\Phi^*/\gamma)^2$ (with linear-throttle convention) admits two solutions over a γ -interval, colliding at a fold γ_F where $\psi'_c(\Phi_F) = 1/\gamma_F^2$. The local normal form on the 1D center manifold is the standard fold

$$\dot{z} = \alpha(\gamma - \gamma_F) - az^2,$$

with $a, \alpha > 0$ in the supercritical case.

Why KROM fails. Two equilibrium branches coexist in $\gamma > \gamma_F$ with L^2 -distance $\sim \sqrt{\gamma - \gamma_F}/\sqrt{a}$. KROM can in principle parametrize each branch separately, but cannot continue between them because the Gauss–Newton iteration is singular at γ_F (the linearization has a zero eigenvalue with eigenvector ϕ^c in its kernel). Pseudo-arclength continuation, which is the standard remedy, is not part of KROM as currently formulated.

6.5 Codim-1: pitchfork on rotation-fixed slice

The MG annulus has $O(2)$ rotation symmetry $\theta \mapsto \theta + \theta_0$. On a rotation-fixed slice (e.g., requiring g to be of pure $\sin(\theta)$ form), the stall bifurcation reduces to a pitchfork in the stall amplitude $A_1 \in \mathbb{R}_{\geq 0}$:

$$\dot{A}_1 = \alpha(\gamma_P - \gamma)A_1 - g_3 A_1^3.$$

For $\gamma < \gamma_P$ there are three equilibria: $A_1 = 0$ (unstable) and $A_1 = \pm \sqrt{\alpha(\gamma_P - \gamma)/g_3}$ (stable, mirror images).

Why KROM fails. Three sheets coexist for $\gamma < \gamma_P$. Snapshots from one stable branch span only that branch; including both branches gives a span containing u_+ , u_- , hence trivially $u_b = (u_+ + u_-)/2$, i.e., the unstable equilibrium. Linear combinations $\beta_+ u_+ + \beta_- u_-$ with $\beta_+ + \beta_- \neq 1$ lie off-manifold. Failure mode (F1) of Section 8.

6.6 Codim-2: Bogdanov–Takens

The Bogdanov–Takens point of MG sits at the intersection of the fold and the stall Hopf in the (γ, B) plane: a parameter pair where the linearization has a double zero eigenvalue with one Jordan block. The unfolding generates fold curves, Hopf curves, and a homoclinic curve emanating from the BT point.

Why KROM fails. The local invariant set near the BT point includes equilibria, limit cycles, and homoclinic orbits, the last of which has no formal Taylor normal form (it is a global object). KROM’s framework has no mechanism for representing a homoclinic orbit as a snapshot.

6.7 Comments

Moore–Greitzer is uniquely valuable as a benchmark because all five of these bifurcations live in the same physical model, with the same compressor characteristic, the same domain, and the same numerical pipeline. Each bifurcation can be studied in isolation by choosing the parameter region appropriately, and combinations can be studied by choosing the codim-2 locus. This makes MG the natural test bed for any branch-aware extension of KROM.

7 Catalogue, part II: PDE benchmarks beyond Moore–Greitzer

The MG model gives an exhaustive set of *local* bifurcations on a single PDE–ODE system, but it does not cover several global and symmetry-driven phenomena that are central to PDE bifurcation theory. We complete the picture with seven additional benchmarks, each chosen to expose a specific feature of the obstruction.

For each benchmark, we follow the same template: equation $\rightarrow$ linear stability $\rightarrow$ bifurcating object $\rightarrow$ why KROM fails (with the failure mode labeled). Three of the benchmarks (Bratu, Swift–Hohenberg, Kuramoto–Sivashinsky) admit explicit *quantitative lower bounds* on the snapshot-bias term, which we state as propositions.

7.1 Bratu–Gelfand: the textbook fold

Equation. The Bratu–Gelfand problem on the unit interval is the semilinear elliptic boundary-value problem

$$-u''(x) = \lambda e^{u(x)}, \quad x \in (0, 1), \quad u(0) = u(1) = 0. \quad (10)$$

Linear stability and bifurcating object. For each λ there are zero, one, or two classical solutions: there exists $\lambda^* \approx 3.51383$ such that for $\lambda < \lambda^*$ there are exactly two solutions (lower-branch $u_-^*(\lambda)$ and upper-branch $u_+^*(\lambda)$) which collide at the fold λ^* and disappear for $\lambda > \lambda^*$. The two branches form a single connected curve in the $(\lambda, \|u\|_\infty)$ plane with a turning point at λ^* . The critical eigenmode is $\phi^c(x) \propto 1/\cosh^2(\theta^*(x - 1/2)/2)$, where $\theta^* \approx 2.39936$ is the merging root of the equation $\sqrt{2\lambda^*} \cosh(\theta/4) = \theta$.

The local fold normal form on the 1D center manifold is $\dot{z} = \alpha(\lambda^* - \lambda) - az^2$ with fold coefficient

$$a = \frac{1}{2} \int_0^1 e^{u_b^*(x)} (\phi^c(x))^3 dx > 0.$$

Near the fold, the Crandall–Rabinowitz parametrization gives

$$u_\pm^*(\lambda) = u_b^* + \frac{\pm 1}{\sqrt{|a|}} \sqrt{\lambda^* - \lambda} \phi^c + O(\lambda^* - \lambda).$$

Why KROM fails. The two branches are at L^2 -distance $\text{dist}(\mathcal{M}_+, \mathcal{M}_-)(\lambda) = 2|a|^{-1/2} \sqrt{\lambda^* - \lambda} \cdot \|\phi^c\|_{L^2} + O(\lambda^* - \lambda)$. A KROM snapshot library on the lower branch consists of functions of the form $u_b^* - |a|^{-1/2} \sqrt{\lambda^* - \lambda} \phi^c + \dots$. The span contains u_b^* and ϕ^c in the limit of refinement, so by sign-flipping it could in principle reach the upper branch. But the PDE constraint at λ does not select the right sign: the Gauss–Newton iteration on a lower-branch initialization will descend back to the lower branch, since the Newton system is precisely the one whose null space contains ϕ^c . Failure mode (F1) plus Gauss–Newton singularity at the fold.

Proposition 7.1 (Bratu lower bound). *Let $\mathcal{M}_-$ denote the closed lower branch of the Bratu problem on $(0, 1)$ truncated at $\lambda \leq \lambda^* - \epsilon_0$ for some $\epsilon_0 > 0$, and let $u^* \in \mathcal{M}_+$ at parameter $\lambda^* - \delta$ with $\delta < \epsilon_0$. Then for any choice of N snapshots in $\mathcal{M}_-$,*

$$\text{dist}(u^*, \mathcal{H}_{K_N})_{L^2(0,1)} \geq \frac{2}{\sqrt{|a|}} \sqrt{\delta} \|\phi^c\|_{L^2} - C \delta,$$

where a is the fold coefficient and C is a universal constant.

Proof sketch. The orthogonal projection of u^* onto $\mathcal{H}_{K_N}$ is constrained to have ϕ^c -component along the lower-branch direction, since every u_i does. The upper-branch target requires the opposite sign of the same eigenmode, with magnitude $|a|^{-1/2} \sqrt{\delta}$. The lower bound follows by the Pythagorean theorem after subtracting the (bounded) $O(\delta)$ remainder. $\square$

7.2 Swift–Hohenberg: continuous-symmetry pitchfork and snaking

Equation. The Swift–Hohenberg equation in 1D is

$$\partial_t u = ru - (1 + \partial_x^2)^2 u + bu^2 - u^3, \quad (11)$$

on a periodic domain of length $L \geq 2\pi$, with r the bifurcation parameter and b the quadratic-coupling parameter (relevant to localized snaking).

Linear stability and bifurcating object. At $r^* = 0$, all modes with wavenumber $k = 1$ become marginally stable, with critical eigenfunction $\phi^c(x) = \cos(x)$ (or any phase rotation thereof, by translation symmetry). For $r > 0$, a non-trivial branch of stationary stripe solutions $u(x) = A(r) \cos(x + \phi)$ emerges from $u = 0$, with amplitude $A(r) = 2\sqrt{r/3} + O(r)$ given by the cubic Stuart–Landau equation. The translation phase $\phi \in S^1$ is a free parameter due to the $x \mapsto x + x_0$ symmetry.

Why KROM fails (F2). The bifurcating set is not a point or a finite collection of points, but a *circle* parametrized by the phase ϕ . The bifurcating manifold is

$$\mathcal{M}_+(r) = \{A(r) \cos(x + \phi) : \phi \in S^1\} \subset L^2(0, 2\pi),$$

a non-flat 1-sphere of radius $A(r)\sqrt{\pi}$ inside a 2D linear span (spanned by $\cos x$ and $\sin x$). KROM's snapshot span is a linear subspace, hence contains the convex hull of $\mathcal{M}_+(r)$ (a 2D disk), not the circle itself. Linear combinations of stripe snapshots produce stripes of varying amplitude, including the trivial $u = 0$, but none of them satisfy the PDE at $r > r^*$.

Proposition 7.2 (Swift–Hohenberg lower bound). *Let $\mathcal{M}_+(r) \subset L^2(0, 2\pi)$ be the post-pitchfork stripe family at $r > 0$. Then for any N snapshots from $\mathcal{M}_+(r)$ at finitely many phases $\phi_1, \dots, \phi_N$, there exists $\phi^* \in S^1$ such that the corresponding stripe $u^*(x) = A(r) \cos(x + \phi^*)$ satisfies*

$$\text{dist}(u^*, \mathcal{H}_{K_N})_{L^2} \geq A(r) \cdot \sqrt{\pi} \cdot \frac{1}{2} \sin\left(\frac{\pi}{N}\right).$$

In particular, the lower bound scales as $A(r)/N$ for large N , and is bounded below by a positive constant uniformly in N for any fixed phase resolution.

Proof sketch. By a pigeonhole argument, any N phases on S^1 leave a gap of at least π/N . A stripe at the midpoint of the largest gap is at L^2 -distance $\geq A(r)\sqrt{\pi} \sin(\pi/(2N))$ from the nearest snapshot. Linear combinations within $\mathcal{H}_{K_N}$ project this distance onto a 2D subspace, halving the bound. $\square$

This is the prototypical (F2) failure: the bifurcating set has non-flat geometry inside its own linear span, and no finite-dimensional linear approximation of a sphere is good in the worst case.

Snaking. For $b > b_c \approx 1.7$, Swift–Hohenberg additionally exhibits localized states that exist on an exponentially-thin parameter window around a Maxwell point, with a saddle-node ladder of widening localized states (the snaking branches). This is a cascade of (F1)-type folds embedded inside the (F2) framework, and is the cleanest PDE example of a fold cascade.

7.3 Diffusive Brusselator: Turing instability

Equation. The Brusselator with diffusion is

$$\begin{cases} \partial_t u = D_u \partial_x^2 u + a - (b+1)u + u^2 v, \\ \partial_t v = D_v \partial_x^2 v + bu - u^2 v, \end{cases} \quad (12)$$

on a periodic domain, with $a, b > 0$ and $D_u, D_v > 0$.

Linear stability and bifurcating object. The homogeneous fixed point is $(u^*, v^*) = (a, b/a)$. Linearizing and Fourier-transforming yields a 2×2 matrix at each wavenumber k . The Turing onset is at

$$b_T = \left(1 + a\sqrt{D_u/D_v}\right)^2, \quad k_T^2 = a/\sqrt{D_u D_v},$$

beyond which a stationary spatially periodic pattern bifurcates from the homogeneous state. There is also a Hopf bifurcation at $b_H = 1 + a^2$. When $D_u = D_v$, the Turing and Hopf onsets coincide, giving a Turing–Hopf codim-2 organizing center.

Why KROM fails (F2). The Turing pattern has translation symmetry, so the bifurcating set is again a circle parametrized by spatial phase. Same (F2) obstruction as Swift–Hohenberg. The codim-2 Turing–Hopf adds a temporal phase, making the local invariant set a 2-torus, which exacerbates (F2).

7.4 Complex Ginzburg–Landau: plane-wave instabilities and defect chaos

Equation. The complex Ginzburg–Landau equation is

$$\partial_t A = A + (1 + i\alpha)\Delta A - (1 + i\beta) \|A\|^2 A, \quad (13)$$

for $A : \Omega \times \mathbb{R} \rightarrow \mathbb{C}$, with parameters $(\alpha, \beta) \in \mathbb{R}^2$.

Plane-wave family. For each $q \in \mathbb{R}$ in a band, there is a plane-wave solution

$$A_q(x, t) = \sqrt{1 - q^2} \exp(iq \cdot x - i\omega(q)t), \quad \omega(q) = \beta + (\alpha - \beta)q^2.$$

Bifurcations. The Benjamin–Feir line $1 + \alpha\beta = 0$ separates parameter regions where plane waves are stable (above) from regions where they are unstable to long-wave perturbations (below). Within the BF-stable region, there is the Eckhaus boundary $q_E^2 = (1 + \alpha\beta)/(3 + 2\alpha^2 + \alpha\beta)$ separating stable from sideband-unstable plane waves. Below the BF line, defect chaos sets in around $1 + \alpha\beta \approx -0.5$.

Why KROM fails. Plane waves at different q are orthogonal in L^2 for q in a discrete grid, so no N -dimensional linear span captures the continuum (F2). Below the BF line, defect chaos involves topological defects (zeros of A around which $\arg A$ winds by $\pm 2\pi$). *The winding number is not preserved by linear combination:* a sum $A_1 + A_2$ of two unit-winding configurations may have winding number 0, ± 1 , ± 2 , depending on relative phase. This is a genuinely topological obstruction (failure mode (F3) with a topological flavor) that cannot be cured by any choice of linear basis.

7.5 Kuramoto–Sivashinsky: the cascade

Equation. The Kuramoto–Sivashinsky equation is

$$\partial_t u + u\partial_x u + \partial_x^2 u + \nu\partial_x^4 u = 0, \quad (14)$$

on a periodic domain of length L with $\nu > 0$. Rescaling: fix $\nu = 1$ and vary L .

Linear stability and the cascade. Linearizing around $u = 0$ gives dispersion $\sigma(k) = k^2 - k^4$, maximized at $k = 1/\sqrt{2}$. Discrete wavenumbers $k_n = 2\pi n/L$ become unstable in succession as L grows:

- For $L < 2\pi$, no unstable modes; trivial state stable.
- At $L = 2\pi$, mode $k_1 = 1$ becomes unstable: pitchfork bifurcation to a stationary cellular state (similar to Swift–Hohenberg but with odd symmetry).
- As L grows past $L_c = 2\pi\sqrt{2}$, more modes become unstable; the cellular state itself loses stability via secondary bifurcations (Hopf, period-doubling, etc.).
- For $L \approx 30$, the system reaches genuine spatiotemporal chaos.

Why KROM fails (F3).

Proposition 7.3 (KS dimension growth). *Let $\mathcal{M}(L)$ denote the inertial manifold of (14) on the periodic domain of length L . Then there exist constants $c_1, c_2 > 0$ such that*

$$c_1 L \leq \dim \mathcal{M}(L) \leq c_2 L$$

for all sufficiently large L .

This is a classical result (Foias–Sell–Temam upper bound, Pomeau–Pumir numerical lower bound). It implies that $\dim \mathcal{H}_{K_N} \geq \dim \mathcal{M}(L)$, and so $N \geq c_1 L$ snapshots are required *just* to make the snapshot bias finite. KS’s snapshot bias scales linearly with the parameter, which is the precise quantitative form of the qualitative statement that “the cascade is fatal”.

Comparison to Moore–Greitzer. MG has finitely many bifurcations on a compact parameter set; the inertial manifold has bounded dimension; the failure of KROM is local in parameter space but globally containable by patching. KS has infinitely many bifurcations as $L \rightarrow \infty$ and the inertial manifold dimension is unbounded; the failure is global, and no patching-based remedy can recover sub-exponential N -width decay. KS is the worst-case benchmark: not because any individual bifurcation is hard, but because the cumulative effect of the cascade defeats reduction.

7.6 Rayleigh–Bénard convection

Equation. Boussinesq Navier–Stokes with thermal coupling on a horizontal slab heated from below:

$$\begin{aligned} \partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u} &= -\nabla p + \text{Pr} \Delta \mathbf{u} + \text{Pr} \text{Ra} T \hat{\mathbf{z}}, \\ \partial_t T + (\mathbf{u} \cdot \nabla) T &= \Delta T, \\ \nabla \cdot \mathbf{u} &= 0, \end{aligned} \tag{15}$$

with no-slip / fixed-temperature boundary conditions at top and bottom and periodic in horizontal directions.

Bifurcation structure. The conduction state $\mathbf{u} = 0, T = 1 - z$ loses stability at $\text{Ra}_c \approx 1707.76$ (rigid–rigid), giving a primary pitchfork to convective rolls (with horizontal translation symmetry, hence (F2)). At higher Rayleigh number, secondary instabilities lead to skewed-varicose, oscillatory, knot, and finally turbulent states. The full bifurcation diagram is famously rich: the parameter region $\text{Ra} \in (\text{Ra}_c, 10^5)$ alone contains dozens of identified instabilities and hosts a Bogdanov–Takens-like organizing center (skewed-varicose).

Why KROM fails. (F2) on the primary pitchfork (continuous translation symmetry of the rolls). (F1) plus (F2) on each secondary instability. (F3) at high Rayleigh number where extensive chaos sets in, with attractor dimension growing as $\text{Ra}^{2/9}$ (Constantin–Foias–Manley–Temam).

7.7 Nonlinear Schrödinger in a double-well potential

Equation. The Gross–Pitaevskii / NLS equation with external potential is

$$i \partial_t \psi = -\Delta \psi + V(x) \psi + g \|\psi\|^2 \psi, \tag{16}$$

with $V(x)$ a symmetric double-well potential and $g \in \mathbb{R}$ the nonlinearity coefficient (focusing or defocusing). Stationary states satisfy $-\Delta \psi + V \psi + g \|\psi\|^2 \psi = \mu \psi$.

Bifurcation structure. For the linear problem ($g = 0$), the ground state is symmetric (even). For g above a critical value g^* (in the focusing case), the symmetric state loses stability via a $\mathbb{Z}_2$ -symmetry-breaking pitchfork to two asymmetric states localized in either well. This is the cleanest discrete-symmetry pitchfork in PDE form: no continuous-symmetry circle, just two isolated post-bifurcation solutions related by the well-exchange symmetry.

Why KROM fails (F1). Same as MG pitchfork on rotation-fixed slice: snapshots from the symmetric branch span only that branch; including both asymmetric branches gives a span containing the symmetric one trivially, but linear combinations weighted differently lie off-manifold. (F1) without the additional geometric complication of (F2).

7.8 Extra: Allen–Cahn / Cahn–Hilliard with parameter

The Allen–Cahn equation $\partial_t u = \Delta u + u - u^3$ on a parameter-dependent domain or with a parameter-dependent potential exhibits multiple coexisting metastable states (multiple wells of the underlying free energy). Each parameter region gives a different combinatorial structure of stable equilibria, with bifurcations between regions corresponding to fold or pitchfork events on individual states. (F1)-type failure throughout, with the additional difficulty that the number of branches can grow combinatorially with system size in 2D and 3D.

8 Three universal failure modes

The catalogue of Sections 6–7 suggests that KROM’s failures fall into three universal classes, characterized by what kind of geometric object the bifurcation introduces. We now formalize these classes; they are independent in the sense that defeating one does not help with the others.

8.1 (F1) Branch separation

Two or more disjoint branches coexist at the same parameter value, with L^2 -distance bounded below by an unfolding scale.

Mathematical signature. For some $\delta > 0$ and at parameter $\mu = \mu_b + s$ with $0 < s < \delta$, the solution manifold satisfies

$$\mathcal{M}(\mu) = \mathcal{M}_1(\mu) \cup \mathcal{M}_2(\mu) \cup \cdots, \quad \text{dist}(\mathcal{M}_i, \mathcal{M}_j)(\mu) \geq c\sqrt{s}.$$

KROM failure. Snapshots from $\mathcal{M}_1$ cannot reach $\mathcal{M}_2$ by linear combination. The snapshot-bias term in (6) is bounded below by the inter-branch distance, independent of N .

Examples. Bratu fold, $\mathbb{Z}_2$ -pitchfork in MG, double-well NLS, snaking ladders in Swift–Hohenberg, transcritical and pitchfork normal forms generally.

Right substitute. A learned center-manifold reduction onto the relevant scalar (fold, $\dot{z} = \alpha s - az^2$) or scalar cubic (pitchfork, $\dot{z} = \alpha sz - g_3 z^3$) normal form. Pseudo-arclength continuation of the snapshot library through the fold.

8.2 (F2) Continuous-symmetry curvature

The bifurcating set is a non-flat submanifold of state space (a circle, torus, or higher group orbit) due to a continuous symmetry of the PDE.

Mathematical signature. The PDE has a continuous symmetry group G (translation, rotation, gauge), and the bifurcating set $\mathcal{M}_+(\mu)$ is an orbit of G inside an isotypic component of G . Linear combinations of G -orbit elements lie in the convex hull of the orbit, which is a higher-dimensional set than the orbit itself.

KROM failure. The snapshot span contains the convex hull, not the orbit. The minimum-norm element of the snapshot span on the constraint set generically lies in the interior of the convex hull, off-orbit, off-manifold, and not satisfying the PDE.

Examples. Swift–Hohenberg stripes, Brusselator Turing patterns, Rayleigh–Bénard rolls, complex Ginzburg–Landau plane waves, MG rotating-stall circle (before phase pinning).

Right substitute. A symmetry-respecting kernel that pre-quotients the orbit. The orbit becomes a single point in the quotient space, and linear approximation in the quotient is well-posed. Equivalently: an encoder that includes the moduli of the symmetry-equivariant Fourier projections (e.g., $\|\Pi_1(g)\|^2$ for the MG first harmonic) as features.

8.3 (F3) Topological / unbounded-dimension growth

The number or dimension of branches grows without bound in the parameter, due to a cascade of bifurcations or topological defect proliferation.

Mathematical signature. The dimension of the inertial manifold, or the count of distinct branches, grows at least linearly with the parameter region considered:

$$\dim \mathcal{M}(L) \geq c_1 L \quad \text{or} \quad \#\text{branches}(L) \geq c_1 L.$$

KROM failure. No fixed- N snapshot library can resolve the full structure for sufficiently large L . The required N grows at least linearly with L , defeating the entire point of reduction.

Examples. Kuramoto–Sivashinsky cascade, complex Ginzburg–Landau defect chaos, Rayleigh–Bénard at high Rayleigh number, period-doubling cascades to chaos.

Right substitute. Adaptive kernels with online snapshot enrichment, multiscale / hierarchical decompositions of the inertial manifold, or acceptance that the regime is fundamentally non-reducible.

8.4 Catalogue mapping

We summarize which benchmark exhibits which failure modes in Table 2.

Benchmark	(F1) Branch separation	(F2) Symmetry curvature	(F3) Dim. growth
MG stall Hopf	yes (cycle vs. eq)	yes (rotation circle)	no
MG fold	yes	no	no
MG pitchfork (slice)	yes	no	no
MG double-Hopf	yes	yes	no
MG Bogdanov–Takens	yes (incl. homoclinic)	no	no
Bratu–Gelfand	yes	no	no
Swift–Hohenberg primary	no	yes	no
Swift–Hohenberg snaking	yes (cascade of folds)	yes	partial
Brusselator Turing	no	yes	no
Brusselator Turing–Hopf	yes	yes	no
Complex GL plane waves	no	yes (continuum)	no
Complex GL defect chaos	yes (topological)	yes	yes
Kuramoto–Sivashinsky	yes (at each step)	yes	yes
Rayleigh–Bénard primary	no	yes	no
Rayleigh–Bénard high Ra	yes	yes	yes
Double-well NLS	yes	no	no

Table 2: Failure-mode catalogue. KS and complex-GL defect chaos are the only benchmarks exhibiting (F3) cleanly; all others have (F1) or (F2) or both.

8.5 Where MG fits

The Moore–Greitzer treatment of Section 6 covers (F1) cleanly across multiple bifurcation types, and the codim-2 cases where two (F1)-type bifurcations interact. It does not cover (F2) genuinely (since the rotation symmetry was broken by phase pinning in our reductions) and does not cover (F3) at all. The PDEs of Section 7 fill out the picture: Bratu and double-well NLS for clean (F1); Swift–Hohenberg, Brusselator, and RBC for (F2) with codim-2 examples; KS and CGL phase turbulence for (F3). The three families are largely independent in the sense that defeating (F1) does not help with (F2) or (F3), and vice versa.

9 POD versus KROM at bifurcations: same blind spot, different failure modes

A natural question that arises once the failure-mode catalogue of Section 8 is in hand is whether POD — the workhorse linear ROM that predates KROM by decades — suffers from the same blind spots, and if so, whether for the same reasons. The answer is more nuanced than a simple yes or no: POD and KROM share the same fundamental approximation-theoretic obstruction at bifurcations, because both are at heart linear-subspace methods on the solution manifold. But mechanically they differ, and the mechanical differences mean the two methods fail in operationally different ways and have different secondary vulnerabilities. We work through the comparison carefully.

9.1 How POD-Galerkin actually operates

We begin by laying out POD-Galerkin’s mechanics in enough detail to see clearly where it differs from KROM. POD-Galerkin proceeds in three distinct stages.

Stage 1: snapshot SVD and basis selection. Given a snapshot library $\{u_i\}_{i=1}^N$ where $u_i = u(\cdot, t_i)$ or $u_i = u(\cdot; \mu_i)$, assemble the snapshot matrix $U \in \mathbb{R}^{n \times N}$ where each column is a discretized snapshot on a grid of n spatial degrees of freedom. Compute the (truncated) singular value decomposition

$$U = \Phi \Sigma V^\top, \quad \Phi \in \mathbb{R}^{n \times r}, \quad \Sigma \in \mathbb{R}^{r \times r}, \quad V \in \mathbb{R}^{N \times r},$$

retaining the leading $r \ll N$ left-singular vectors as the POD basis $\{\phi_1, \dots, \phi_r\}$. The number r is chosen so that the captured energy

$$\frac{\sum_{i=1}^r \sigma_i^2}{\sum_{i=1}^N \sigma_i^2} \geq 1 - \varepsilon$$

exceeds a target threshold (typically $1 - \varepsilon = 99\%$ or higher).

Stage 2: Galerkin projection of the PDE. Substitute the ansatz $u(t, x) \approx \sum_{i=1}^r a_i(t) \phi_i(x)$ into the time-dependent PDE and project the residual onto each basis function:

$$\langle \phi_j, \partial_t u - \mathcal{P}(u) \rangle = 0, \quad j = 1, \dots, r.$$

This produces a reduced ODE system for the coefficient vector $\mathbf{a}(t) = (a_1, \dots, a_r)^\top$:

$$\dot{\mathbf{a}} = \mathbf{F}(\mathbf{a}), \quad \mathbf{F}_j(\mathbf{a}) = \langle \phi_j, \mathcal{P}(\sum_i a_i \phi_i) \rangle. \quad (17)$$

For *linear* PDEs the right-hand side $\mathbf{F}(\mathbf{a})$ can be precomputed offline as a constant linear operator on $\mathbf{a}$, and online cost is $\mathcal{O}(r^2)$ per time step. For *nonlinear* PDEs, evaluating $\mathbf{F}(\mathbf{a})$ requires reconstructing the full state $\sum_i a_i \phi_i$ on the high-dimensional grid, evaluating the nonlinearity, and projecting back, which is $\mathcal{O}(n)$ per time step — defeating the reduction. This motivates hyperreduction (DEIM, GNAT, LSPG, missing-point estimation), which we discuss in Section 9.5.

Stage 3: time integration of the reduced ODE. Integrate (17) with a standard ODE solver (Runge–Kutta, BDF, etc.) from initial condition $\mathbf{a}(0) = (\langle \phi_1, u_0 \rangle, \dots, \langle \phi_r, u_0 \rangle)^\top$. Reconstruct the full state at any desired time as $u(t, x) \approx \sum_i a_i(t) \phi_i(x)$.

Summary. POD-Galerkin produces a low-dimensional dynamical system that mimics the full PDE on the leading-energy subspace. The reduced model is a closed ODE system that one integrates forward in time.

9.2 How KROM operates by contrast

KROM, as developed in Sections 2–3, does not derive a reduced ODE. It uses the snapshot span as the RKHS for an optimal-recovery problem, and at each query (each new parameter μ , or each new time step in a time-marching scheme) it solves the full PDE collocation system in the snapshot subspace via Gauss–Newton.

The mechanical contrast is sharp:

- POD pushes a low-dimensional reduced ODE forward in time. At any moment the reduced state is $\mathbf{a}(t) \in \mathbb{R}^r$ and the dynamics are governed by the projected operator $\mathbf{F}$ alone.

- KROM solves the full PDE constraints (interior, boundary, initial) at every time step or parameter query. The constraints involve the full differential operator $\mathcal{P}$ acting on a function in $\mathcal{H}_{K_N}$, evaluated at collocation sites, and Gauss–Newton iterations are run until the constraints are satisfied to tolerance.

This means: *POD has its own internal dynamics on the reduced subspace; KROM does not.* At each query, KROM solves a constrained recovery problem that always touches back to the full PDE. POD, once the projection is committed to, evolves the reduced ODE as a self-contained dynamical system whose trajectory is determined by the projected operator, not by re-solving against the PDE at each step.

9.3 What both methods share: the Kolmogorov N -width barrier

Despite the mechanical differences, POD and KROM are both *linear-subspace methods at the level of approximation theory*. POD approximates the solution by a linear combination of r leading singular modes; KROM approximates the solution by an element of the linear span of the snapshots (with a data-dependent inner product). In both cases, the hypothesis space is an r - or N -dimensional linear subspace of the ambient function space.

This is a crucial observation, because the Kolmogorov N -width

$$d_N(\mathcal{M}; V) = \inf_{\substack{W \subset V \\ \dim W = N}} \sup_{u \in \mathcal{M}} \inf_{v \in W} \|u - v\|_V$$

is a property of the solution manifold $\mathcal{M}$ alone, independent of the algorithm used to construct the approximating subspace. *Any* linear-subspace method — POD, reduced-basis, KROM with the empirical kernel, or hand-designed modal expansion — is bounded below by $d_N(\mathcal{M}; V)$ in worst-case error.

Proposition 9.1 (Shared Kolmogorov barrier). *Let $\mathcal{M} \subset V$ be the solution manifold of a parametrized PDE, and let W be any N -dimensional linear subspace of V . Let $u^* \in \mathcal{M}$ be the worst-case target. Then for any method whose approximation $u^\diamond$ lies in W ,*

$$\|u^* - u^\diamond\|_V \geq d_N(\mathcal{M}; V).$$

In particular, both POD’s leading-energy subspace and KROM’s snapshot span are N -dimensional linear subspaces, so both inherit this lower bound.

The consequence at bifurcations is immediate. If $\mathcal{M}$ is multi-sheeted with branches separated by $\delta = \text{dist}_V(\mathcal{M}_1, \mathcal{M}_2) > 0$, then by the elementary argument used throughout this report,

$$d_N(\mathcal{M}; V) \geq \frac{\delta}{2}$$

for all sufficiently large N . Both POD and KROM are bounded below by $\delta/2$ in worst-case error; neither can approximate the post-bifurcation manifold to accuracy better than the inter-branch distance.

Failure-mode coverage. The (F1), (F2), and (F3) failure modes of Section 8 apply identically to both methods:

- (F1) Branch separation: A pitchfork in Allen–Cahn produces two separated branches; both POD and KROM are bounded below by $\text{dist}(u_+, u_-)$ for any snapshot library on a single branch.

- (F2) Continuous-symmetry curvature: A Swift–Hohenberg stripe family is a circle in a 2D linear span; both POD’s $\text{span}\{\cos x, \sin x\}$ and KROM’s $\text{span}\{u_1, \dots, u_N\}$ contain the convex hull (a disk), not the circle, with the same L^2 -distance bound from the worst stripe.
- (F3) Unbounded-dimension growth: Kuramoto–Sivashinsky’s inertial manifold dimension grows linearly in L ; both POD and KROM need $r, N \gtrsim cL$ to make the worst-case error finite, defeating reduction in both cases.

The classes of systems where POD and KROM fail at the level of approximation theory are identical. A pitchfork is fatal to POD just as it is to KROM; a rotating-stall limit cycle is bounded away from the equilibrium branch in H^1 regardless of which linear-subspace method you choose.

9.4 Where they diverge: three mechanical differences

Although the approximation-theoretic floor is the same, the way the floor is reached — and what additional failures occur on top of it — differs in three concrete ways.

9.4.1 Divergence 1: stability of the reduced model

POD-Galerkin reduced models are well-known to suffer from stability issues even on problems where the underlying PDE is perfectly stable. The reason is that Galerkin projection does not preserve all spectral properties of the original operator. Concretely, suppose the linearized PDE has spectrum $\sigma(L) \subset \mathbb{C}_-$ (all eigenvalues in the open left half-plane, so the equilibrium is stable). Then the Galerkin-projected operator $L_r = \Phi^\top L \Phi$ has spectrum $\sigma(L_r)$ that depends on how well Φ resolves the slow stable directions versus the fast stable directions. For convection-dominated or transport-aligned problems, the leading-energy POD modes capture the dominant transport but miss small-scale dissipative modes that are crucial for stability; the projected operator can have eigenvalues on or even to the right of the imaginary axis even when the truth is uniformly stable.

This is the well-documented stability fragility of POD-Galerkin and motivates much of the literature on stabilization (penalty methods, Petrov–Galerkin variants like LSPG, structure-preserving projections, closure modeling).

Why this matters at bifurcations. At a bifurcation parameter μ_b , the truth has, by definition, an eigenvalue on the imaginary axis. Small perturbations of the spectrum — from Galerkin truncation error, from numerical stiffness, or from inadequate snapshot diversity — can shift the projected critical eigenvalue to either side of the axis, causing the reduced model to predict the *wrong side* of the bifurcation. A POD-Galerkin reduction trained on snapshots from $\mu < \mu_H$ may, at $\mu = \mu_H + \delta$ for small $\delta > 0$, predict either a still-stable equilibrium (missing the bifurcation entirely) or an unstable equilibrium with a spurious limit cycle whose amplitude does not match the true Hopf scaling. This is a confidently-wrong reduced model: the user obtains a numerical answer that looks reasonable but disagrees qualitatively with the truth.

KROM does not have this issue. KROM does not integrate a reduced ODE forward in time. There is no truncated reduced operator whose spectrum can drift away from the truth’s. At each query, KROM solves a constrained recovery problem against the full PDE, and the answer either satisfies the PDE constraints to tolerance (in which case it is geometrically close to the truth, modulo the snapshot-bias term) or the Gauss–Newton iteration fails to converge (in which case the user is alerted that no $\mathcal{H}_{K_N}$ -element satisfies the constraints). KROM cannot produce a confidently-wrong

reduced model in the POD sense; it can only produce a no-answer or a geometrically-bounded approximation.

9.4.2 Divergence 2: the Gauss-Newton singularity at folds

The flip side of the comparison: KROM uses Gauss-Newton iteration to solve the nonlinear collocation system. The Newton system at iteration k is

$$\mathcal{L}(\mu) \delta u = -[\mathcal{P}(u^{(k)}; \mu) - f],$$

where $\mathcal{L}(\mu)$ is the linearization of $\mathcal{P}$ at $u^{(k)}$. At a fold parameter μ_F , by definition, $\mathcal{L}(\mu_F)$ has the critical eigenmode ϕ^c in its kernel, so the Newton system is singular and the iteration cannot make progress.

This is a generic issue at any fold, and the standard remedy is pseudo-arclength continuation: parameterize the branch by an arclength variable s rather than the parameter μ , and treat μ itself as an unknown along with u . The Newton system on (u, μ) is regular at the fold even though the system on u alone is singular. KROM as currently formulated does *not* implement pseudo-arclength continuation; the user supplies μ as an input, and the solver attempts to recover u at that fixed μ .

POD’s situation at a fold. POD-Galerkin, having reduced to a low-dimensional ODE system, treats fold continuation in the reduced coordinates by exactly the same pseudo-arclength method one would use for any low-dimensional ODE. The reduced system $\dot{\mathbf{a}} = \mathbf{F}(\mathbf{a}; \mu)$ at equilibrium $\mathbf{F}(\mathbf{a}^*; \mu) = 0$ has a singular Jacobian at $\mu = \mu_F$, but the augmented system $(\mathbf{a}, \mu)$ continued by arclength is regular. This is a textbook problem in numerical bifurcation analysis (AUTO, MatCont, COCO are mature tools for it). So in principle, *POD has a more natural mechanism for navigating folds than KROM does*, provided the basis Φ has been chosen rich enough to span both branches near the fold.

The caveat “provided the basis is chosen well” is doing real work here. If the snapshot library is restricted to one side of the fold, the reduced ODE will not have the second branch as an equilibrium at all, and continuation cannot manufacture one. So POD’s advantage at folds is conditional on having representative data, exactly as KROM’s recovery is conditional on having a representative snapshot library.

9.4.3 Divergence 3: handling of non-smooth solution structure

A separate divergence, independent of bifurcations but relevant to the catalogue: KROM with the empirical kernel handles non-smooth solution structure (shocks, sharp interfaces, discontinuous coefficients) substantially better than POD. The reason is that POD’s leading singular modes are global L^2 -functions, and the truncation error in the leading r modes typically smooths sharp features because high-frequency content is concentrated in the discarded tail of the SVD. KROM’s empirical kernel, by contrast, inherits whatever non-smoothness lives in the snapshots: if every snapshot has a shock at the same location, then every linear combination has the shock there too, and the Gibbs-like oscillations that plague stationary kernels (Matérn, Gaussian) are absent.

This divergence is the empirical advantage that the original KROM paper demonstrates on viscous Burgers, Allen–Cahn, and discontinuous-coefficient Darcy flow. It is not a bifurcation-related advantage — it shows up away from any bifurcation parameter — but it is one of the main practical reasons to prefer KROM over POD on problems with sharp gradients.

9.5 Hyperreduction: DEIM, GNAT, LSPG

For completeness we comment briefly on the hyperreduction methods that improve POD-Galerkin’s nonlinear-term cost. The discrete empirical interpolation method (DEIM) approximates the nonlinear term $\mathcal{N}(u)$ as

$$\mathcal{N}(u) \approx \Phi_{\mathcal{N}}(P^\top \Phi_{\mathcal{N}})^{-1} P^\top \mathcal{N}(u),$$

where $\Phi_{\mathcal{N}}$ is a low-rank basis for the nonlinear-term snapshot ensemble and P is a binary selection matrix that picks $p \ll n$ rows. This makes evaluating $\mathcal{N}(u)$ cost $\mathcal{O}(p)$ instead of $\mathcal{O}(n)$ and recovers the reduction’s online efficiency. Variants such as Q-DEIM (pivoted-QR for point selection), GNAT (Gauss–Newton for approximated tensors), and LSPG (least-squares Petrov–Galerkin) refine the procedure. Missing-point estimation and gappy POD use related ideas to handle partial observations or to choose interpolation points greedily.

Hyperreduction does not change the bifurcation picture. All hyperreduction methods are still linear-subspace methods at the level of approximation theory: they project the solution and the nonlinear term onto leading-singular-mode subspaces of dimensions r and p respectively. Proposition 9.1 applies unchanged. A pitchfork or a Hopf bifurcation defeats DEIM, GNAT, and LSPG to the same degree as plain POD-Galerkin, and for the same geometric reason. Hyperreduction makes the linear-subspace method affordable on nonlinear PDEs; it does not make the linear subspace any less linear.

9.6 The genuinely-different ROM family: nonlinear manifold methods

The genuinely different approaches that escape the linear-subspace assumption are those that build a *nonlinear* manifold approximation. These include:

- *Quadratic-manifold POD* (Barnett–Farhat, Geelen–Willcox): augment the linear subspace by an explicit quadratic correction $u \approx \sum a_i \phi_i + \sum a_i a_j \phi_{ij}$, which can curve away from the linear span. This is sufficient for some (F2)-type problems with mild curvature but does not address (F1) or (F3).
- *Convolutional / variational autoencoder ROMs* (Kashima, Lee–Carlberg, Fresca–Manzoni, Conti–Gobat–Fresca–Manzoni–Frangi): learn an encoder/decoder pair from data with a neural network. These can in principle represent multi-sheeted manifolds but do so at the cost of non-convex training, no convergence guarantees, and large data requirements.
- *Implicit / continuous neural representations* (CROM, neural-ODE-based ROMs): represent the reduced state as a learned mapping rather than a basis expansion.
- *Kernel-conjugacy / normal-form ROMs*: the parallel-project pipeline this report describes for Moore–Greitzer. Learn a nonlinear conjugacy onto a structured low-dimensional ODE whose normal form is determined by the bifurcation type.

The kernel-conjugacy approach, developed in Section 10, is the natural extension of KROM into the nonlinear-manifold family while preserving KROM’s RKHS / minimum-norm / sparse-Cholesky machinery.

9.7 Side-by-side comparison

Table 3 summarizes the comparison.

Aspect	POD-Galerkin	KROM
Approximation hypothesis	r -dim leading-energy sub-space	N -dim snapshot span (RKHS)
Online procedure	Integrate reduced ODE forward	Solve constrained recovery
PDE constraints satisfied	Approximately (Galerkin)	Exactly at collocation sites
Boundary conditions	Approximately enforced	Exactly by construction
Nonlinearity handling	Hyperreduction (DEIM, GNAT)	Gauss–Newton in RKHS
Online cost (post-hyperred.)	$\mathcal{O}(r^2 + p)$	$\mathcal{O}(N_{\text{data}} \log^{2d} N_{\text{data}})$
Subject to $d_N(\mathcal{M}; V)$ lower bound	Yes	Yes
Fails at (F1) branch separation	Yes	Yes
Fails at (F2) symmetry curvature	Yes	Yes
Fails at (F3) unbounded growth	Yes	Yes
Reduced-operator stability fragility	Yes (well-known)	No
Can produce confidently-wrong answer	Yes	No
Fold continuation in reduced coords.	Available (pseudo-arclength)	Singular Newton, no continuation
Native handling of shocks, discontinuities	Poor (smoothing)	Good (snapshot-inherited)
Maturity of toolkit	Decades, well-developed	Recent, evolving

Table 3: POD-Galerkin and KROM compared. The first block describes mechanics, the second block lists the shared approximation-theoretic limits, and the third block lists the mechanical divergences. The two methods share the same Kolmogorov- N -width-bounded failure profile at bifurcations but have complementary operational personalities everywhere else.

9.8 The honest summary

POD and KROM are siblings in the linear-subspace family of ROMs. They share the same fundamental approximation-theoretic obstruction at bifurcations — both are bounded below by the Kolmogorov N -width of the solution manifold, and both inherit the (F1), (F2), (F3) failure modes identically. The classes of systems where the approximation-theoretic failure occurs are identical for both methods.

Their mechanical personalities, however, differ:

- POD-Galerkin is more flexible in the reduced coordinates — it has a self-contained reduced ODE on which standard pseudo-arclength continuation works at folds, and it benefits from decades of mature numerical-bifurcation tooling.
- KROM is more honest about the full PDE constraints — it cannot produce a confidently-wrong reduced model in the POD-Galerkin stability sense, and it handles shocks and sharp gradients natively because the empirical kernel inherits the non-smoothness of the snapshots.
- POD has the stability fragility of projected operators; KROM has the Gauss–Newton singularity at folds.

Each method’s distinctive mechanical advantage corresponds to a distinctive mechanical disadvantage of the other.

The proper extension is the same for both. The crucial point for this report is that *the proper extension of either POD or KROM to handle bifurcations is the same*: replace the linear subspace by a learned nonlinear chart onto a normal form. Whether you start from POD-Galerkin or from KROM, the destination is the kernel-conjugacy / normal-form pipeline of Section 10. The (F1)/(F2)/(F3) taxonomy is method-agnostic; the resolution is method-agnostic; and the unified branch-aware framework of Section 12 can be built on either scaffolding. The report’s emphasis on KROM rather than POD is a matter of starting point, not of which method “has the bifurcation problem” — both do, equally, and both are cured by the same nonlinear-conjugacy substitute.

10 The right object: nonlinear conjugacy onto a normal form

If linear approximation of the solution manifold breaks at a bifurcation, the right replacement is a *nonlinear* conjugacy onto a model dynamical system whose topological structure already encodes the bifurcation. This idea has a long history in dynamical systems theory under the name of normal-form theory; the new ingredient is to learn the conjugacy from data using kernel methods, exploiting the same RKHS / minimum-norm / computational-graph-completion machinery that powers KROM.

10.1 Center manifold and Poincaré normal form

The classical setup is as follows. Consider an evolution system

$$\dot{X} = G(X; \mu),$$

on a Banach space $\mathcal{X}$, with a smooth equilibrium branch $X^*(\mu)$. Shifting $y = X - X^*(\mu)$ gives $\dot{y} = \tilde{G}(y; \mu)$ with $\tilde{G}(0; \mu) = 0$, and the linearization $A(\mu) = D\tilde{G}(0; \mu)$ has spectrum $\sigma(A(\mu))$. Suppose at $\mu = \mu_b$ a critical part of the spectrum crosses the imaginary axis transversally, with the rest bounded away from $i\mathbb{R}$.

The center manifold theorem produces, locally near $(0, \mu_b)$, a smooth invariant manifold W^c of dimension equal to the number of critical eigenvalues. The dynamics restricted to W^c are a low-dimensional ODE that captures the entire local bifurcation behavior. The Poincaré normal form theorem further conjugates these reduced dynamics to a model system in canonical form, depending on the codimension and resonance structure.

For example, for a generic codim-1 Hopf bifurcation, the truncated normal form in complex coordinates is

$$\dot{z} = (\rho(\mu) + i\omega(\mu))z + \beta(\mu) \|z\|^2 z + \delta(\mu) \|z\|^2 Jz + \mathcal{O}(\|z\|^5). \quad (18)$$

For a fold, the normal form is the scalar

$$\dot{z} = \alpha(\mu - \mu_b) - az^2. \quad (19)$$

For a pitchfork in a $\mathbb{Z}_2$ -symmetric setting, it is the scalar cubic

$$\dot{z} = \alpha(\mu - \mu_b)z - g_3 z^3. \quad (20)$$

Each of these is a 1- or 2-dimensional ODE with the bifurcation structure built in.

10.2 Why this is the right replacement for snapshot reduction

A normal-form-based reduction differs from a snapshot subspace in three structural ways:

1. *It is a low-dimensional ODE, not a linear subspace.* The reduced state $z \in \mathbb{R}^k$ is not a coordinate in a linear subspace of $\mathcal{X}$; it is an intrinsic coordinate on the center manifold, and the relationship $X \mapsto z$ is genuinely nonlinear.
2. *Its bifurcation structure is built in.* The unfolding parameter $\rho(\mu)$ encodes the eigenvalue crossing as a parameter of the model, not as a pattern that the method has to discover. The bifurcation diagram of (18), (19), or (20) is exact.
3. *Locality is dynamical, not spatial.* The reduction is local in phase space, in a neighborhood of the equilibrium branch in $\mathcal{X} \times \mathcal{D}$.

10.3 Learning the conjugacy with kernels: the GP framework

A separate paper in this project (Hamzi–Owhadi and collaborators on *Cole–Hopf and Poincaré normal forms via Gaussian processes*) shows that one can learn the conjugating map $\Phi : \mathcal{X} \times \mathcal{D} \rightarrow \mathbb{R}^k$ from trajectory data by kernel ridge regression in an RKHS. The architecture has three blocks:

- an *encoder* Φ in an RKHS $\mathcal{H}_\Phi$ that maps the high-dimensional state to the latent variable z ;
- a *decoder* Ψ in an RKHS $\mathcal{H}_\Psi$ that reconstructs the state from z ;
- a *coefficient block* that fits the parametric coefficient maps of the chosen normal form.

The training loss is a computational-graph-completion (CGC) objective that combines:

$$\begin{aligned}
 L_{\text{conj}} &= \sum_i w_i \left\| D\Phi(y_i, \mu_i) \tilde{G}(y_i; \mu_i) - N(\Phi(y_i, \mu_i), \mu_i) \right\|^2 && \text{(infinitesimal conjugacy)} \\
 L_{\text{rec}} &= \sum_i w_i \left\| \Psi(\Phi(y_i, \mu_i), \mu_i) - y_i \right\|^2 && \text{(round-trip reconstruction)} \\
 L_{\text{gauge}} &= \text{(gauge fixing for the normal-form ambiguity)} \\
 \|\Phi\|_{\mathcal{H}_\Phi}^2, \|\Psi\|_{\mathcal{H}_\Psi}^2 &= \text{(RKHS regularization)}.
 \end{aligned}$$

The optimization is solved by an alternating scheme: encode the data, fit the normal-form coefficients by ridge regression, fit the decoder by kernel ridge, then refit the encoder by L-BFGS, and iterate. The minimum-RKHS-norm regularization is the same Owhadi-style optimal-recovery principle that drives KROM, applied at the level of the conjugating map rather than the solution itself.

10.4 Codim-1 normal forms, side by side

For each of the codim-1 normal forms relevant to our catalogue, the architecture instantiates with a different latent ODE:

- *Fold.* $z \in \mathbb{R}$, $\dot{z} = \alpha(\mu - \mu_b) - az^2$. Two unknowns a, α . Locality weights $\propto \exp(-(z/r_{\text{loc}})^2)$.
- *Pitchfork* ($\mathbb{Z}_2$). $z \in \mathbb{R}$, $\dot{z} = \alpha(\mu - \mu_b)z - g_3 z^3$. Two unknowns α, g_3 . Locality weights centered at $z = 0$, with reflection-symmetric loss.
- *Hopf.* $z \in \mathbb{C}$, equation (18). Four parametric unknowns $\rho, \omega, \beta, \delta$ as functions of μ . Polar-decomposed loss.

- *Transcritical.* $z \in \mathbb{R}$, $\dot{z} = \alpha(\mu - \mu_b)z - az^2$. Two unknowns. Need to fit two intersecting branches; locality weights centered on each branch.

For each, the encoder and decoder kernels can be the same ones used for Hopf (Matérn with polynomial features); only the latent ODE and the coefficient-block parameterization change.

10.5 Codim-2 normal forms

For codim-2, the latent dimension is 2 (Bautin, Bogdanov–Takens, fold-Hopf, cusp) or 4 (double-Hopf), and the coefficient block has more unknowns:

- *Bautin (degenerate Hopf).* Hopf form (18) with $\beta(\mu_b) = 0$ at the Bautin point; need quintic correction $|z|^4 z$.
- *Bogdanov–Takens.* 2D normal form $\dot{u} = v$, $\dot{v} = \mu_1 + \mu_2 v + au^2 + buv$. Six unknowns, with a homoclinic curve emanating from the BT point that is *not* part of the Taylor normal form.
- *Cusp.* 1D, $\dot{z} = \mu_1 + \mu_2 z + cz^3$. Three unknowns, hysteresis loop.
- *Double-Hopf.* 4D. Two unfolding parameters, two frequencies, four cubic coefficients.

The kernel-conjugacy framework extends to all of these by changing the latent ODE, but the homoclinic in the BT case requires a separate treatment (no Taylor reduction for global objects).

11 The Moore–Greitzer extension: KROM’s natural successor in the bifurcation regime

We now describe in detail the application of the kernel-conjugacy machinery to Moore–Greitzer, which is the parallel research thread in this project. The exposition makes the connection to KROM precise: where KROM offers a linear-snapshot RKHS reduction that fails at the bifurcation, the Moore–Greitzer pipeline offers a nonlinear-conjugacy RKHS reduction that handles the bifurcation by construction.

11.1 State space and target

The PDE state is $X = (\Phi, \Psi, g) \in \mathcal{X}$ (Section 6). For each of the five MG bifurcations, the target reduced model is a different normal form:

- Codim-1 stall Hopf $\rightarrow$ (18).
- Codim-2 mixed surge–stall $\rightarrow$ double-Hopf.
- Codim-1 fold $\rightarrow$ (19).
- Codim-1 pitchfork $\rightarrow$ (20).
- Codim-2 Bogdanov–Takens $\rightarrow$ BT 2D normal form.

We focus the exposition on the codim-1 stall Hopf since it is the simplest and the most physically central; the others are obtained by changing the latent ODE.

11.2 Encoder kernel

The encoder $\Phi : \mathcal{X} \times \mathbb{R} \rightarrow \mathbb{R}^2$ is built from a function-space kernel on the spatial slot of the state. A practical choice mixes a Matérn-5/2 kernel on the $L^2(S^1)$ part of g with low-order polynomial features in the rotational invariants

$$\Pi_k(g) := \frac{1}{2\pi} \int_0^{2\pi} g(\theta) e^{ik\theta} d\theta, \quad k = 1, 2, 3,$$

so that the encoder RKHS contains exactly the features that appear in the local Hopf reduction: the first-harmonic complex amplitude, its square, and the radial intensity $\|\Pi_1\|^2$. This is the (F2)-respecting structure of Section 8: the rotation symmetry of the annulus is built in as a feature of the kernel, so the bifurcating circle becomes a single point in the encoder space.

11.3 Decoder kernel

The decoder $\Psi : \mathbb{R}^2 \times \mathbb{R} \rightarrow \mathcal{X}$ is operator-valued, with separable kernel

$$\mathbf{K}_\Psi((z, \gamma), (z', \gamma')) = k_z(z, z') k_\gamma(\gamma, \gamma') T,$$

where $T = \text{diag}(1, 1, T_g)$ acts on $\mathcal{X}$ and T_g is the heat semigroup on S^1 with a learned smoothing scale. This is the function-valued analog of the Matérn-on- $\mathbb{R}^2$ kernel ridge in finite-dimensional Brusselator and Goodwin experiments.

11.4 CGC loss

The training loss is the PDE-aware version of the GP-paper objective described in Section 10.3. The infinitesimal conjugacy term reads

$$L_{\text{conj}} = \sum_i w_i \|D\Phi(X_i, \gamma_i)[G(X_i; \gamma_i)] - N(\Phi(X_i, \gamma_i), \gamma_i)\|^2,$$

where $G(X_i; \gamma_i)$ is the right-hand side of (9) evaluated at the training trajectory point and the Fréchet derivative $D\Phi(X_i, \gamma_i)$ acts on a function-valued increment. Together with the round-trip reconstruction loss, gauge fixing, and RKHS regularization, this reproduces in function space the architecture used previously for Brusselator and Goodwin.

Locality weights:

$$w_i \propto \exp\left(-\left(\frac{\|y_i\|}{r_{\text{loc}}}\right)^2\right) \exp\left(-\left(\frac{\gamma_i - \gamma_H}{\sigma_{\text{loc}}}\right)^2\right),$$

concentrating the loss in a neighborhood of the equilibrium branch in $\mathcal{X} \times \mathbb{R}$, where center-manifold theory says the conjugacy is valid.

11.5 Alternating optimization

The training scheme is the alternating procedure used elsewhere in the project:

1. with current Φ , compute latent variables $z_i = \Phi(X_i, \gamma_i)$ and latent velocities $\dot{z}_i = D\Phi(X_i, \gamma_i)G(X_i; \gamma_i)$;
2. project onto radial and angular components, fit $\rho(\gamma), \omega(\gamma), \beta(\gamma), \delta(\gamma)$ by weighted ridge regression with structured parametric forms $\beta(\gamma) = \beta_0 + \beta_1\rho(\gamma)$, $\delta(\gamma) = \delta_0 + \delta_1\rho(\gamma)$;
3. fit the decoder by operator-valued kernel ridge;
4. refit the encoder by L-BFGS on the full CGC objective;
5. iterate until convergence.

11.6 Why this works where KROM fails

The Moore–Greitzer pipeline addresses every one of the obstructions identified in Section 4 and Section 8:

- *(F1) addressed by nonlinear chart.* The encoder is genuinely nonlinear in the state; it contains polynomial features in the Fourier projections $\Pi_k(g)$. The hypothesis space is no longer a linear subspace of $\mathcal{X}$, so it is not subject to Kolmogorov N -width lower bounds across multi-sheeted manifolds.
- *(F2) addressed by symmetry-respecting kernel.* The polynomial features in $\|\Pi_1(g)\|^2$ are rotation-invariant; the rotating-stall circle is a single point in the encoder space.
- *Bifurcation structure built in.* The latent dynamics are of the form $\dot{z} = (\rho(\gamma) + i\omega(\gamma))z + \beta(\gamma)\|z\|^2 z + \dots$. The bifurcation point is recovered by solving $\rho(\gamma_H) = 0$; the criticality is read off from the sign of $\beta(\gamma_H)$. There is no need for the snapshot library to contain limit-cycle states: the limit cycle for $\gamma < \gamma_H$ is generated by integrating the latent ODE and decoded back to $\mathcal{X}$ by Ψ .
- *Locality in phase space.* The training data is sampled in a neighborhood of the equilibrium branch in $\mathcal{X} \times \mathbb{R}$, exactly where center-manifold theory says the conjugacy is valid.
- *Sparse-Cholesky machinery still applicable.* For the encoder and decoder kernel matrices on training data, the Schäfer–Owhadi sparse-Cholesky approximation is still a valid acceleration. The sparse-Cholesky ingredient of KROM is preserved; only the kernel and the recovery target change.

11.7 Validation strategy

Validation against the same kinds of metrics used in the Brusselator and Goodwin experiments, with PDE-specific extensions:

- *Held-out PDE rollout.* Integrate the latent normal form, decode through Ψ , and compare against a high-fidelity numerical solution of (9) in H^1 .
- *Reconstruction.* Compare the reconstructed $g(\theta, t)$ heatmap against the true heatmap.
- *Bifurcation point recovery.* Compare the learned γ_H (from $\rho(\gamma_H) = 0$) and the learned frequency ω_H against numerical eigenvalue continuation on the linearization of (9).
- *First Lyapunov coefficient.* Compare the learned cubic coefficient $\beta_0 = \beta(\gamma_H)$ against the classical adjoint-eigenvector calculation.
- *Mesh-refinement study.* Vary the spatial quadrature on θ and verify that γ_H , ω_H , and β_0 converge in the mesh, providing evidence that the learned reduction is a property of the PDE rather than of the discretization.
- *Cubic-coefficient sensitivity.* For Moore–Greitzer specifically, the cubic coefficient β is amplitude-sensitive. Sampling at radius comparable to the predicted limit-cycle saturation is essential for robust identification.

11.8 Honest assessment

The Moore–Greitzer kernel-conjugacy pipeline has been validated cleanly on the codim-1 stall Hopf with mesh refinement. The codim-2 mixed surge–stall double-Hopf is a partially successful experiment: the stall channel identifies cleanly, but the surge channel remains under-excited at typical sampling protocols, with cubic cross-coupling coefficients erratic. The fold, pitchfork, and Bogdanov–Takens cases have been worked out in theory and partially implemented; the BT case requires special handling of the homoclinic that is genuinely outside the Taylor framework.

12 A unified branch-aware framework

KROM and the kernel-conjugacy pipeline are not competitors but complementary tools targeting distinct regimes. We summarize the comparison and outline a unified framework.

12.1 Comparison

	KROM	Kernel pipeline	normal-form
Reduction object	Linear span $\mathcal{H}_{K_N}$ of snapshots	Nonlinear conjugacy onto a normal form	
What is recovered	PDE solution u^*	Local flow up to smooth conjugacy	
Inductive bias	Solution lies in snapshot span	Local dynamics has a known normal form	
Locality	Spatial (screening on Ω)	Dynamical (center manifold near X^*)	
Bifurcation handling	Fails (multi-sheeted manifold)	Built in (target ODE has the bifurcation)	
Genericity	Non-intrusive, parameter-agnostic	Requires equilibrium branch and codimension	
Sparse-Cholesky use	Yes, online-critical	Yes, on encoder/decoder Gram matrices	
Theoretical rate	$h_\Omega^\gamma + d_N(\mathcal{M}; V)$	Center-manifold approximation theorem	
Failure mode coverage	None of (F1), (F2), (F3)	(F1), (F2) at codim-1; partial codim-2; not (F3)	

12.2 The unified pipeline

A unified, branch-aware kernel reduction framework would combine the two pipelines as follows.

1. *Branch detection.* Use eigenvalue continuation along $X^*(\mu)$ to detect candidate bifurcation parameters μ_b and classify them by codimension and normal-form type.
2. *Region partitioning.* Split the parameter set $\mathcal{D}$ into open subsets $\mathcal{D}_k$ separated by detected bifurcations.

3. *Per-region kernel choice.*

- On each $\mathcal{D}_k$ where Assumption 3.3 holds (no bifurcation crossings), use KROM with an empirical kernel built from snapshots local to that region.
 - For (F2)-affected regions, use a symmetry-respecting kernel (rotation- or translation-invariant features) to pre-quotient the symmetry orbit.
4. *Bifurcation patches.* On a neighborhood of each μ_b , switch to the kernel-conjugacy pipeline with target ODE chosen by codimension and normal-form type:
- codim-1: fold, pitchfork, transcritical, Hopf;
 - codim-2: Bautin, Bogdanov–Takens, cusp, fold-Hopf, double-Hopf;
 - global: homoclinic patches require separate treatment (e.g., Lin’s method).
5. *Stitching.* At the boundaries between branches, enforce continuity by sharing kernel hyperparameters and by matching the encoder/decoder on overlap regions, in the spirit of partition-of-unity methods.
6. *Cascade handling.* For (F3) regimes, abandon a fixed- N snapshot library; use online enrichment with target accuracy as a function of L , and accept that the asymptotic complexity is no better than the inertial-manifold dimension.

A natural first step is to implement the branch-detection and codim-1 patch components on the pure-stall Moore–Greitzer experiment, then add fold and pitchfork patches, then Swift–Hohenberg and Bratu as cleaner test problems with closed-form ground truth.

12.3 Open questions

The catalogue and the analysis raise several theoretical questions:

1. *Nonlinear N -widths for multi-sheeted manifolds.* The Kolmogorov N -width is a linear notion. For multi-sheeted manifolds the natural object is a nonlinear or piecewise-linear N -width, where the approximating set is allowed to be a union of K subspaces of dimension N/K rather than a single subspace of dimension N . There is some literature on this (Cohen– DeVore–Petrova–Wojtaszczyk, recent ROM-from-snapshots work), but no clean theory in the kernel / RKHS context yet.
2. *Symmetry-respecting kernels.* For (F2) failures, a continuous-symmetry-respecting kernel would pre-quotient the rotation/translation orbit. This is closely related to invariant kernels in machine learning, but the connection to PDE bifurcation theory has not been worked out.
3. *Center-manifold approximation in RKHS norm.* For all three failure modes, the right local target is a center-manifold reduction. Quantitative approximation theorems for the center manifold in RKHS norm (rather than the more classical C^k or H^s norms) would underpin a unified branch-aware framework.
4. *Sparsity decay for empirical kernels.* The KROM paper proves sparse-Cholesky efficiency for stationary Matérn-type kernels, but the analogous theory for empirical kernels is only sketched. Bifurcation-aware empirical kernels would have additional structure (block-diagonal in branches, off-diagonal coupling near bifurcation points) that may admit sharper sparsity statements.

5. *Global bifurcations.* Homoclinic and heteroclinic bifurcations have no formal Taylor normal form. Existing kernel-conjugacy machinery does not handle them. Lin’s method or boundary-value-problem formulations of the homoclinic orbit are the classical numerical tools; integrating them into a kernel-based framework is open.
6. *Adaptivity for cascade regimes.* For (F3), the right framework is online snapshot enrichment with a target accuracy. The interaction between online enrichment and sparse-Cholesky updates is open.

13 Conclusion

KROM is an elegant and powerful method for kernelized reduced-order modeling of nonlinear PDEs, combining empirical kernels with sparse Cholesky factorization to produce a non-intrusive, near-linear-complexity solver that adapts to problem-specific structure. Its theoretical guarantees rest on a uniform-coercivity / analytic-parameter-dependence assumption that ensures sub-exponential Kolmogorov N -width decay of the snapshot-bias term.

We have argued in detail that this assumption fails by definition at bifurcation points of the solution manifold, and the failure manifests structurally: the empirical-kernel RKHS is a linear span of snapshots, which cannot, by linear combination, reach solutions on a different branch of the manifold. We have characterized which dynamical regimes KROM handles — precisely those where the local normal form has empty critical spectrum, equivalently those where the equilibrium branch is uniformly hyperbolic — and given a graded taxonomy of how violently it fails for each codim-1 and codim-2 normal form. We have instantiated the analysis on the Moore–Greitzer compressor model with its five elementary local bifurcations, and on seven additional canonical PDEs (Bratu, Swift–Hohenberg, Brusselator, complex Ginzburg–Landau, Kuramoto–Sivashinsky, Rayleigh–Bénard, double-well NLS), with explicit quantitative lower bounds on the snapshot-bias term where possible. The catalogue suggests three universal failure modes — (F1) branch separation, (F2) continuous-symmetry curvature, (F3) topological / unbounded-dimension growth — which are largely independent: a method addressing one does not automatically address the others.

The natural successor to KROM in the bifurcation regime is the kernel-based normal-form learning pipeline developed in parallel for the Moore–Greitzer compressor model. Where KROM offers a linear-snapshot reduction, the normal-form pipeline offers a nonlinear-conjugacy reduction onto a low-dimensional bifurcation model, with the bifurcation structure built into the target ODE and locality restored to phase space where it belongs. The same Owhadi-style RKHS minimum-norm regularization that powers KROM is reused at the level of the conjugating map. The two pipelines are complementary, and a unified, branch-aware framework that uses KROM in regions of analytic parameter dependence, switches to symmetry-respecting kernels in (F2)-affected regions, and applies normal-form patches across bifurcations is the natural next step for the project. The catalogue of PDE benchmarks above provides the test cases for each component of the extension, with closed-form ground truth where possible (Bratu, NLS double-well, Swift–Hohenberg) and high-quality numerical ground truth otherwise (Brusselator, KS, RBC, CGL, Moore–Greitzer).

A Reference data for each benchmark

For convenience we collect the basic constants and scalings for each benchmark.

A.1 Bratu–Gelfand on $(0, 1)$

- Fold parameter: $\lambda^* \approx 3.51383$.
- Fold solution: $\|u_b^*\|_\infty \approx 1.18684$.
- Critical mode: $\phi^c \propto 1/\cosh^2(\theta^*(x - 1/2)/2)$ where $\theta^* \approx 2.39936$ is the merging root of $\sqrt{2\lambda^*} \cosh(\theta/4) = \theta$.
- Fold coefficient: $a = \frac{1}{2} \int_0^1 e^{u_b^*} (\phi^c)^3 dx > 0$.
- Branch separation: $\text{dist}(\mathcal{M}_+, \mathcal{M}_-)(\lambda) = 2|a|^{-1/2} \sqrt{\lambda^* - \lambda} \|\phi^c\|_{L^2} + O(\lambda^* - \lambda)$.

A.2 Swift–Hohenberg on $[0, 2\pi)$

- Pitchfork at $r^* = 0$, critical wavenumber $k^* = 1$.
- Critical mode: $\phi^c(x) = \cos x$ (or any phase rotation thereof).
- Cubic normal-form coefficient: $g_3 = 3$ (with the $-u^3$ nonlinearity at face value).
- Bifurcating amplitude: $A(r) = 2\sqrt{r/3} + O(r)$.
- Snaking pinning interval (1D, $b > b_c \approx 1.7$): $r \in (r_-, r_+)$ with width $\sim \exp(-\pi/\sqrt{b - b_c})$, exponentially thin in the cascade parameter.

A.3 Diffusive Brusselator

- Fixed point: $(u^*, v^*) = (a, b/a)$.
- Hopf onset: $b_H = 1 + a^2$, frequency $\omega_H = a$.
- Turing onset: $b_T = (1 + a\sqrt{D_u/D_v})^2$, critical wavenumber $k_T^2 = a/\sqrt{D_u D_v}$.
- Turing–Hopf coincidence: requires $D_u = D_v$ (degenerate; in practice tuned by changing the model).

A.4 Complex Ginzburg–Landau

- Plane-wave family: $A_q = \sqrt{1 - q^2} e^{iq \cdot x - i\omega(q)t}$, $\omega(q) = \beta + (\alpha - \beta)q^2$.
- Benjamin–Feir line: $1 + \alpha\beta = 0$.
- Eckhaus boundary: $q_E^2 = (1 + \alpha\beta)/(3 + 2\alpha^2 + \alpha\beta)$ within the BF-stable region.
- Defect-chaos onset: empirically $1 + \alpha\beta \approx -0.5$ for typical (α, β) .

A.5 Kuramoto–Sivashinsky

- Linear dispersion: $\sigma(k) = k^2 - k^4$, max at $k = 1/\sqrt{2}$.
- Pitchfork onset: $L = 2\pi$ (mode $k_1 = 1$ becomes unstable).
- Transition to chaos: $L \approx 30$.
- Inertial manifold dimension: $\dim \mathcal{M}(L) \asymp L$ for large L .

A.6 Moore–Greitzer (cubic compressor)

- Compressor characteristic: $\psi_c(\Phi) = \psi_0 + H(1 + \frac{3}{2}(\Phi/\Theta - 1) - \frac{1}{2}(\Phi/\Theta - 1)^3)$.
- Stall Hopf: γ_H at $\psi'_c(\Phi^*) =$ critical value depending on viscosity.
- Fold of axisymmetric equilibria (linear throttle, $\psi_0 = 0$): $\gamma_F = 1/\psi'_c(\Phi_F) = 4.9383$, $\Phi_F = 1.5$.
- Fold normal-form coefficients: $a^{\text{theory}} = -\frac{1}{2} \langle p, B(q, q) \rangle = +0.157$, $\alpha^{\text{theory}} = +0.071$.

A.7 Rayleigh–Bénard (rigid–rigid)

- Critical Rayleigh number: $\text{Ra}_c \approx 1707.76$.
- Critical wavenumber: $k_c \approx 3.117$.
- Skewed-varicose instability of straight rolls: $\text{Ra} \sim 2\text{Ra}_c$, depending on Prandtl number.
- Attractor dimension at high Ra : $\dim \mathcal{A} \sim \text{Ra}^{2/9}$ (Constantin–Foias–Manley–Temam upper bound).

A.8 Double-well NLS (focusing, 1D)

- Symmetric ground state stable for $\|\psi\|_{L^2}^2 < N_c$.
- Symmetry-breaking pitchfork at $N = N_c$, with two asymmetric branches for $N > N_c$.
- Critical N_c depends on the inter-well barrier height and spacing; for a Gaussian-bump double well of standard parameters $N_c \approx 0.2$.

B Notation and conventions

- Ω : spatial domain in $\mathbb{R}^n$. $\tilde{\Omega} = [0, T] \times \Omega$.
- $\mathcal{D}$: parameter set. $\mu \in \mathcal{D}$ a generic parameter.
- $\mathcal{M}$: solution manifold $\{u(\mu) : \mu \in \mathcal{D}\}$.
- V : ambient Hilbert space (typically $H_0^s(\Omega)$ for some s).
- K, K_N : kernel; empirical kernel.
- $\mathcal{H}_K, \mathcal{H}_{K_N}$: associated RKHS.
- h_Ω : spatial fill distance of collocation points.
- ρ : sparsity radius for sparse Cholesky.
- N : snapshot count. N_{data} : total collocation count.
- μ_b : bifurcation parameter value. μ_H : Hopf parameter. μ_F : fold. μ_P : pitchfork.
- ϕ^c : critical eigenmode at the bifurcation.
- Φ : encoder. Ψ : decoder.
- z : latent normal-form coordinate.